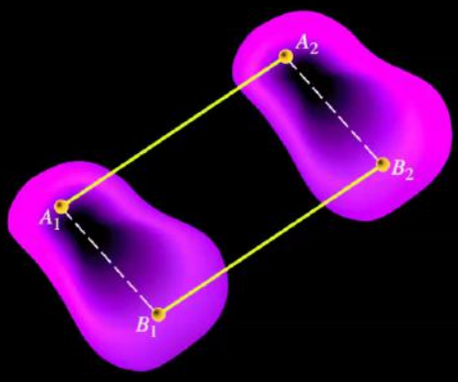
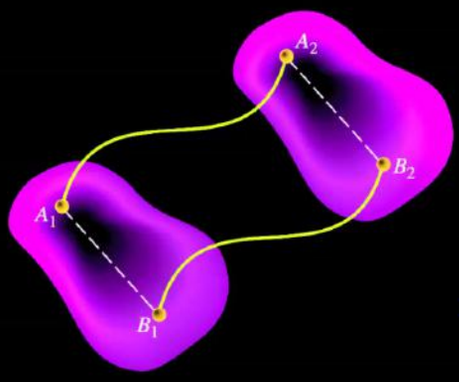
Kinematics Of Rigid Bodies:

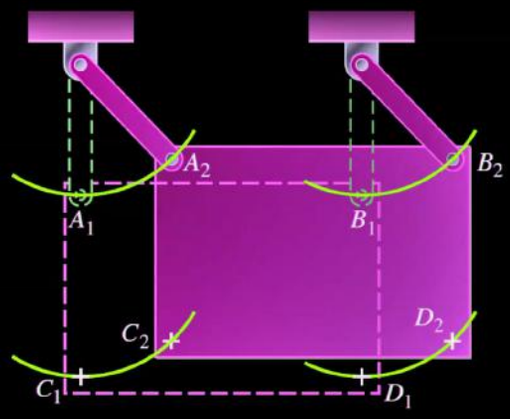
Translation:



A rigid body in rectilinear translation.



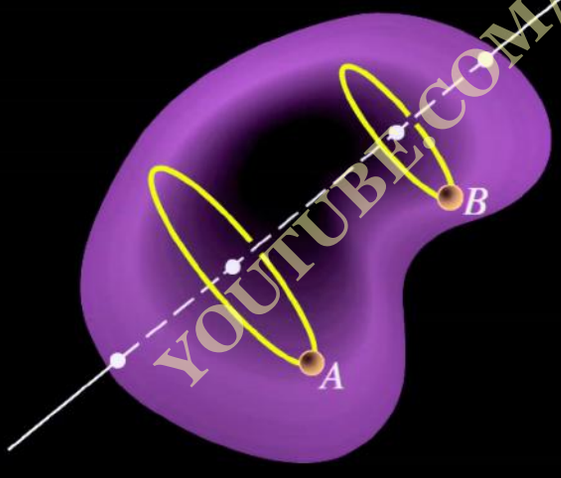
A rigid body in curvilinear translation.



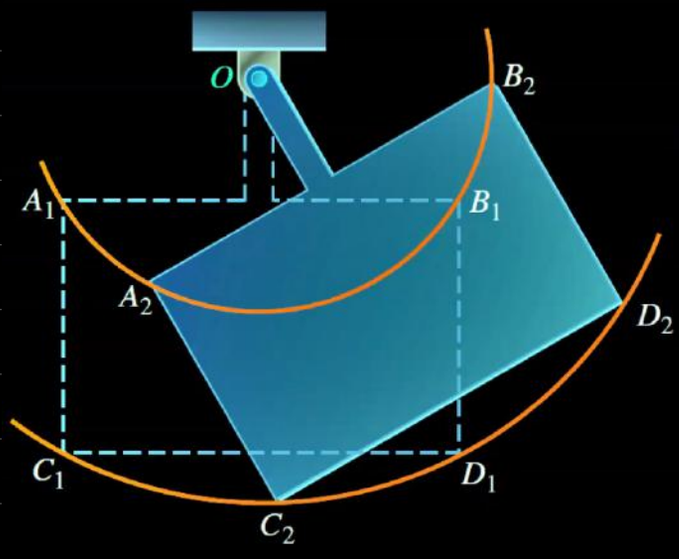
Curvilinear translation

A motion is said to be translation if any straight line inside the body maintains its orientation during the motion. If paths followed by particles during such a motion are straight lines, we call it rectilinear translation and if they are curves, we call it curvilinear translation.

Rotation About A Fixed Axis:



A rigid body rotating about a fixed axis.



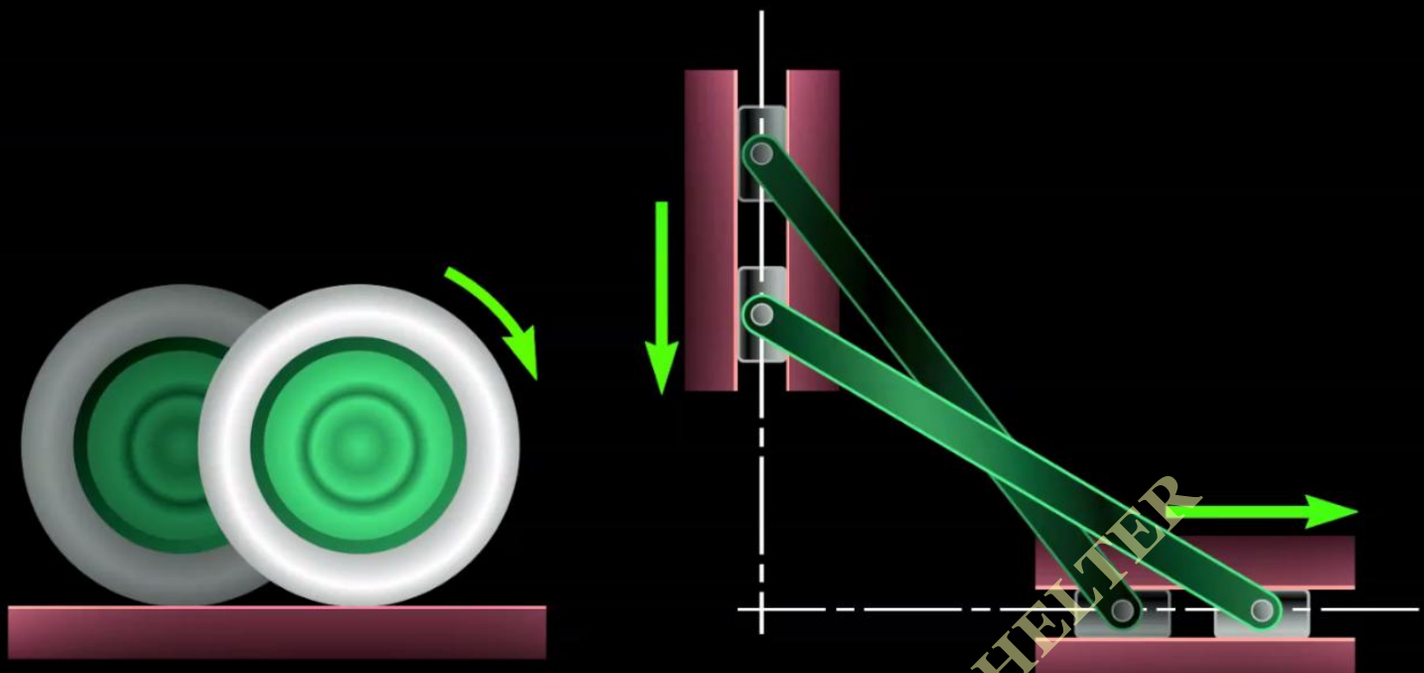
Rotation

In this type of motion, particles of the rigid body move in parallel planes along circles centred on the same fixed axis. This axis is called axis of rotation. The particles of the rigid body lying on the axis of rotation have zero velocity.

→ Make sure to never confuse between curvilinear translation with rotation.

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General Plane Motion:

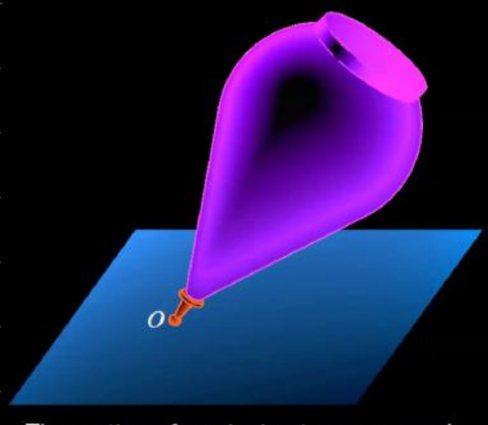


A motion in which all particles of a rigid body move in a plane (though each particle could be moving in its own plane) is called General plane motion. A planar motion which is neither pure translation, nor a fixed axis rotation is usually called General plane motion (Though translation and fixed axis rotation may be considered as special cases of General plane motion).

→ Note that classification of a motion is frame dependent. For example from frame of the axle, a rolling wheel is undergoing fixed axis rotation.

Motion About A Fixed Point:

The three dimensional motion of a rigid body attached at a fixed point O, such as motion of a top on a rough floor is known as motion about a fixed point (In this motion, one of the points on the rigid body maintains its position throughout the motion).



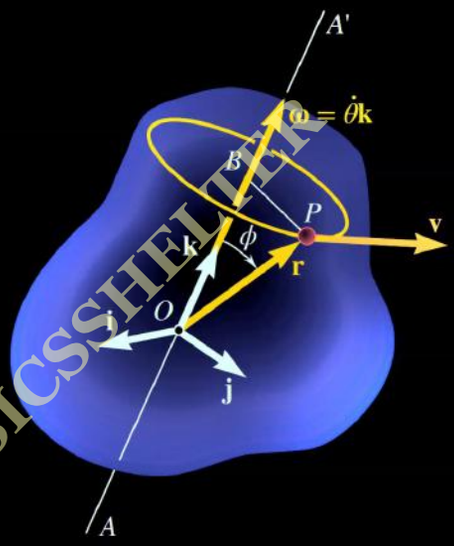
The motion of a spinning top on a rough surface is an example of three-dimensional motion about a fixed point.

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General Motion: Any motion of a rigid body that does not fall in to any of the above categories is called General motion. As an example a spinning top inside a moving vehicle as seen from ground frame will fall in to this category.

Velocity Of An Arbitrary Point In Fixed Axis Rotation:

Consider an arbitrary rigid body undergoing fixed axis rotation as shown in the figure. Here particle P is going in a circle of radius BP and its speed is given as $\omega \times BP = v$ as per our understanding from rotational mechanics. Also recall that we defined angular velocity vector using right hand thumb rule (i.e curl the fingers in the sense of rotation then the thumb gives the direction of $\vec{\omega}$ [Right hand Rule]). Thus $\vec{v} = \vec{\omega} \times \vec{BP}$ (as derived in circular motion). Now let O be an arbitrary point on the axis of rotation such that position vector of P is $\vec{r}$ wrt O. Now we can write $\vec{BP} = \vec{BO} + \vec{OP}$, thus, we can write velocity of P as:



For a rigid body in rotation about a fixed axis, the velocity of a particle is the vector product of the angular velocity of the body and the position vector of the particle.

$$\vec{v} = \vec{\omega} \times (\vec{BO} + \vec{OP}) = \vec{\omega} \times \vec{BO} + \vec{\omega} \times \vec{r}$$

but since $\vec{\omega}$ and $\vec{BO}$ are collinear, $\vec{\omega} \times \vec{BO} = 0$ thus, for any arbitrary point we can write the velocity vector as:

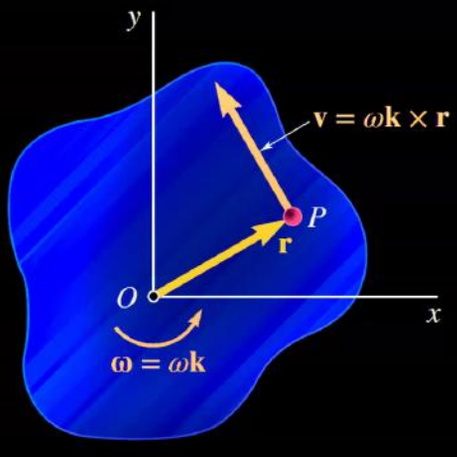
$$\boxed{\vec{v} = \vec{\omega} \times \vec{r}}$$

$\vec{v}$: velocity vector $\vec{r}$: Position vector

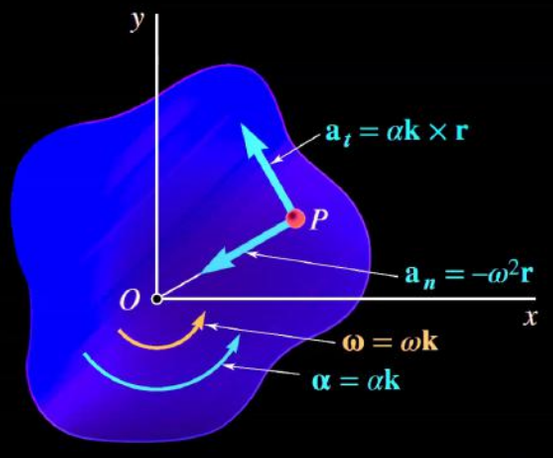
$\vec{\omega}$: Angular velocity vector as given by right hand thumb rule.

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Acceleration Of Arbitrary Point On Rigid Body (Fixed Axis):



For an object undergoing fixed-axis rotation, the velocity of a point *P* equals the vector product of the angular velocity vector and the position vector of *P*. A positive value of the scalar ω corresponds to counterclockwise motion.



For an object undergoing fixed-axis rotation, the acceleration of a point *P* has a tangential component that depends on angular acceleration, as well as a normal component that depends on angular velocity.

In the previous section we derived velocity vector of an arbitrary point. To find acceleration vector, we can simply differentiate the velocity vector thus:

$$\vec{a} = d\vec{v}/dt = d/dt\,[\vec{\omega} \times \vec{r}] = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

Thus $\boxed{\vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v}}$ or

$\boxed{\vec{a} = (\vec{\alpha} \times \vec{r}) + \vec{\omega} \times (\vec{\omega} \times \vec{r})}$ To properly understand the significance of two terms, Suppose we consider rotation of a thin slab about Z axis. Then, $\vec{v} = \omega\hat{k} \times \vec{r}$, now since $\vec{\omega}$ and $\vec{r}$ are mutually $\perp$, $v = \omega r$. Now if we substitute $\vec{\alpha} = \alpha\hat{k}$ and $\vec{\omega} = \omega\hat{k}$ in the expression for acceleration, we get

$$\vec{a} = \alpha\hat{k} \times \vec{r} + (\omega\hat{k} \times (\omega\hat{k} \times \vec{r}))$$

$$\Rightarrow \vec{a} = \alpha\hat{k} \times \vec{r} - \omega^2\vec{r}$$ Now we can resolve $\vec{a}$ in to radial and tangential components.

$$\vec{a}_t = \alpha\hat{k} \times \vec{r} \Rightarrow a_t = \alpha r$$

$$\vec{a}_c = -\omega^2\vec{r} \Rightarrow a_c = \omega^2 r$$

These are the familiar results that we derived in our study of circular motion.

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Equations Defining Rotation Of Rigid Body About Fixed Axis:

The motion of a rigid body about a fixed axis is said to be defined if we know its rotation angle as a function of time i.e $\theta(t)$ is known, however using laws of motion or work energy analysis we typically know $\alpha(\theta)$ or $\alpha(t)$ or $\omega(\theta)$ or $\omega(t)$ and we need differential equations to relate various variables. Such differential equations follow from definitions of kinematic variables α, ω and θ and we derived them in circular motion. Here we state these differential equations for a ready reference:

$$\omega = d\theta/dt \qquad \alpha = d\omega/dt = d^2\theta/dt^2 \qquad \text{and} \qquad \alpha = \omega\, d\omega/d\theta$$

We can now integrate these differential equations appropriately to get the desired results. Two frequently occurring case of fixed axis rotation are:

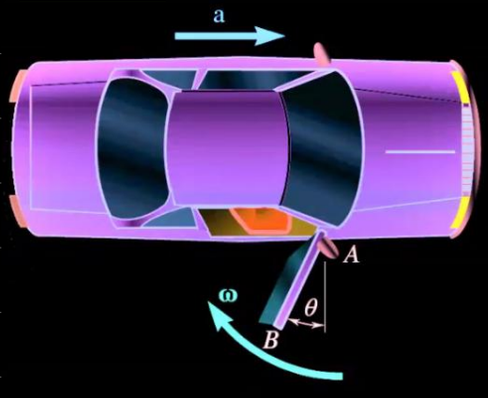
Uniform Rotation: This case refers to the condition where angular acceleration is zero, therefore angular velocity is constant and $\theta(t)$ is given as, $\theta(t) = \theta_0 + \omega t$

Uniformly Accelerated Rotation: In this case, angular acceleration is constant and the equations governing angular kinematics are similar to those governing rectilinear motion. These equations are:

$$\omega = \omega_0 + \alpha t$$
$$\theta = \theta_0 + \omega t + \tfrac{1}{2}\alpha t^2$$
$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

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Example:

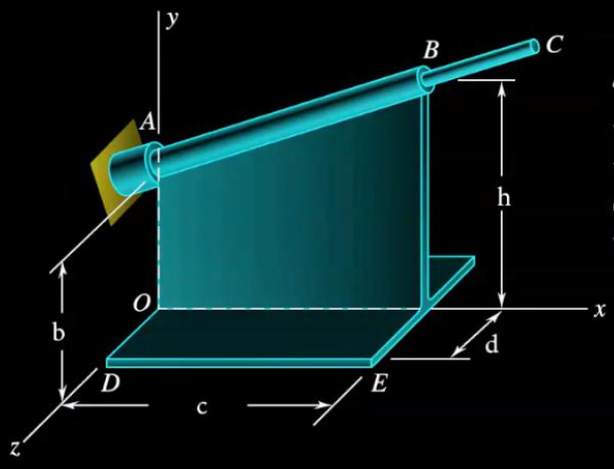


A driver starts his car with the door on the passenger's side wide open ($\theta = 0$). As the car moves forward with constant acceleration, the angular acceleration of the door is $\alpha = 2.5 \cos\theta$, where α is in rad/s^2. Determine the angular velocity of the door as it slams shut ($\theta = 90°$).

Solution: $\alpha = \frac{\omega d\omega}{d\theta} = 2.5\cos\theta \Rightarrow \int_0^{\omega} \omega\, d\omega = \int_0^{\pi/2} 2.5\cos\theta\, d\theta$

$\Rightarrow \frac{\omega^2}{2} = 2.5 \Rightarrow \omega = \sqrt{5}$ rad s^{-1} ≈ 2.4 rad s^{-1} Ans.

EXAMPLE:



The assembly shown rotates about the rod AC. At the instant shown, the assembly has an angular velocity of ω rad/s that is increasing with an angular acceleration of α rad/s^2. Knowing that the y component of the velocity of corner D is negative at this instant in time, determine the velocity and acceleration of corner E.

Solution: We must first carefully decide the direction of $\vec{\omega}$. Recall that direction of $\vec{\omega}$ is given by right hand thumb rule, now since y component of velocity of corner D is negative, $\vec{\omega}$ must be along $\vec{AC}$ ($\vec{\omega}$ along $\vec{CA}$ will make y component of $\vec{v}_D$ positive). Thus,

$$\vec{\omega} = \frac{\omega[c\hat{i} + (h-b)\hat{j}]}{\sqrt{c^2 + (h-b)^2}}$$ and since ω in increasing, $\vec{\alpha}$ must also be along $\vec{\omega}$, thus $\vec{\alpha} = \frac{\alpha\,(c\hat{i} + (h-b)\hat{j})}{\sqrt{c^2 + (h-b)^2}}$

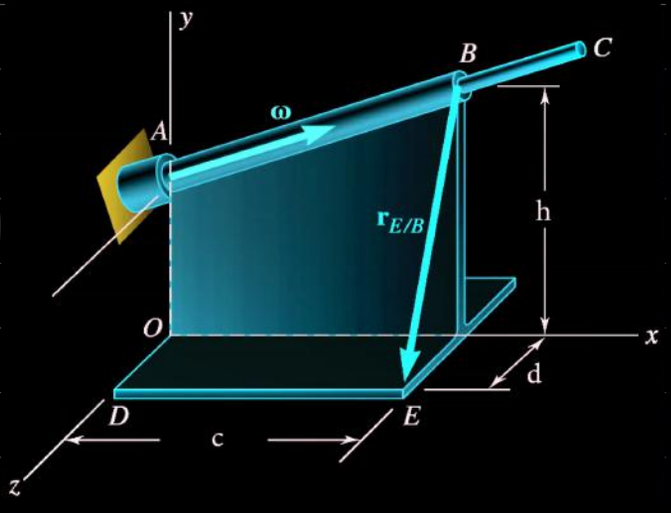
Now let us choose origin at B, then position vector of E is $\vec{r} = -h\hat{j} + d\hat{k}$

Now velocity of $\vec{E} = \vec{\omega} \times \vec{r}$

$\Rightarrow$

$$\vec{v} = \frac{\omega}{\sqrt{c^2 + (h-b)^2}}\left[-ch\hat{k} - cd\hat{j} + (h-b)d\,\hat{i}\right]$$

$$\Rightarrow \vec{v} = \frac{\omega}{\sqrt{c^2 + (h-b)^2}}\left[(h-b)d\,\hat{i} - cd\hat{j} - ch\hat{k}\right]$$ Ans.



Now Tangential acceleration, $\vec{a}_t = \vec{\alpha} \times \vec{r}$, this expression will be similar

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to $\vec{\omega} \times \vec{r}$, thus recognizing this we can directly write:

$$\vec{a}_t = \frac{\alpha}{\sqrt{c^2 + (h-b)^2}} \left[(h-b)d\,\hat{i} - cd\,\hat{j} - ch\,\hat{k} \right] \text{ Ans.}$$

Now centripetal acceleration $\vec{a}_c = \vec{\omega} \times \vec{v}$, we have already found $\vec{v}$

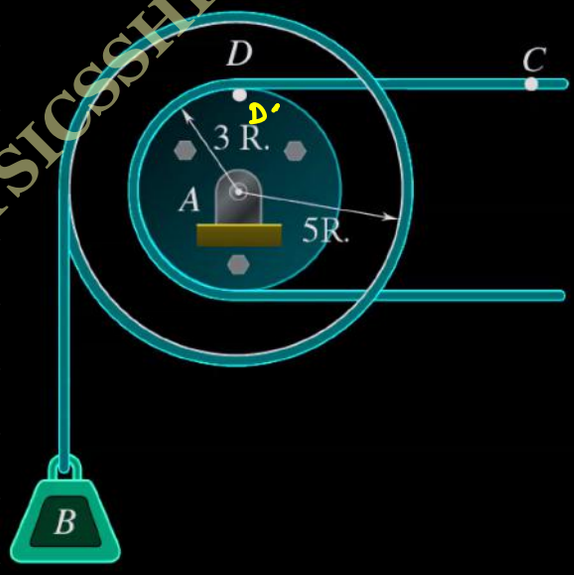
Thus:

$$\vec{a}_c = \frac{\omega^2}{c^2 + (h-b)^2} \left[c\hat{i} + (h-b)\hat{j} \right] \times \left[(h-b)d\hat{i} - cd\hat{j} - ch\hat{k} \right]$$

$$\Rightarrow \vec{a}_c = \frac{\omega^2}{c^2 + (h-b)^2} \left[-ch(h-b)\,\hat{i} + c^2 h\,\hat{j} - d\left((h-b)^2 + c^2\right)\hat{k} \right] \text{ Ans.}$$

MEANING OF NO SLIPPING:

Whenever we say that two contacting particles do not slip wrt each other it means that their speed is same. For instance consider the belt and stepped pulley arrangement shown in adjacent figure, Here particle D of the driving belt is in contact with particle D' of the inner step of the stepped pulley, thus for no slipping $v_D = v_{D'}$ also since this condition will always hold true even if the particles keep changing, so long as slipping does not happen, this means that derivatives of velocity at this contact must also be equal. Therefore no slipping also implies that tangential accelerations must be equal.

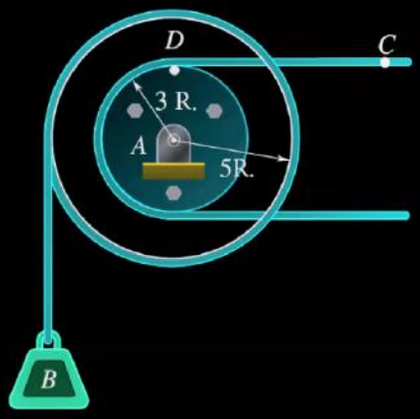


→ No slipping implies equal tangential accelerations

→ No slipping implies equal speeds at every instant at the point of contact.

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EXAMPLE:



Load B is connected to a double pulley by one of the two inextensible cables shown. The motion of the pulley is controlled by cable C, which has a constant acceleration of 9g and an initial velocity of u, both directed to the right. Determine (a) the number of revolutions executed by the pulley in T s, (b) the velocity and change in position of the load B after T s, (c) the acceleration of point D on the rim of the inner pulley at $t = 0$.

Solution: (a) At point D, $a = R\alpha \Rightarrow 9g = 3R\alpha \Rightarrow \alpha = 3g/R$

also $U = \omega R \Rightarrow u = \omega \times 3R \Rightarrow \omega = u/3R$.

Now $\Delta\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ and number of revolutions: $N = \frac{\Delta\theta}{2\pi}$

Thus $N = \left[\frac{u}{3R}T + \frac{3g}{2R}T^2\right].\frac{1}{2\pi}$ Ans.

(b) String length wrapped $= SR \times \Delta\theta = SR\left[\frac{uT}{3R} + \frac{3g}{2R}T^2\right]$

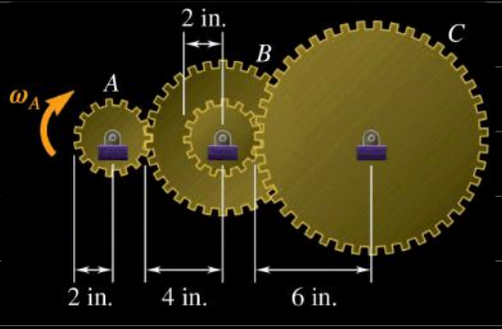
Thus height ascended by the load: $\left[\frac{5}{3}uT + \frac{15}{2}gT^2\right]$ Ans.

(c) Now tangential acceleration of D is same as that of C i.e 9g, centripetal acceleration $= u^2/3R$ thus net acceleration of D:

$\vec{a}_D = 9g\hat{i} - \frac{u^2}{3R}\hat{j}$ Ans.

EXAMPLE:

A gear reduction system consists of three gears A, B, and C. Knowing that gear A rotates clockwise with a constant angular velocity $\omega_A = 600$ rpm, determine (a) the angular velocities of gears B and C, (b) the accelerations of the points on gears B and C which are in contact.



Solution: Due to no slipping, tangential speeds at contact points must be same. Thus $\omega_A r_A = \omega_B r_B$ i.e $600\,rpm \times 2'' = \omega_B \times 4'' \Rightarrow \omega_B = 300$ RPM.

Similarly due to inner gear contact with C: $\omega_B \times 2'' = \omega_C \times 6''$

$\Rightarrow \omega_C = 100$ RPM.

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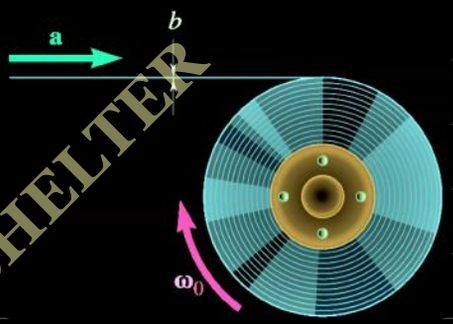
Now acceleration $a_c = \omega^2 r$ thus for inner gear, $a = \omega_B^2 \times 2in$

i.e. $\left(300 \times \frac{2\pi}{60}\right)^2 \times 2in \approx 2 \times 10^3 \ in/sec^2$, Similarly for C,

$a = \left(300 \times \frac{2\pi}{60}\right)^2 \times 6in \approx \frac{2}{3} \times 10^3 \ in/sec^2$ Ans.

EXAMPLE:

Steel tape is being wound onto a spool that rotates with a constant angular velocity ω_0. Denoting by r the radius of the spool and tape at any given time and by b the thickness of the tape, derive an expression for the acceleration of the tape as it approaches the spool.



Solution: Instant velocity of the tape:

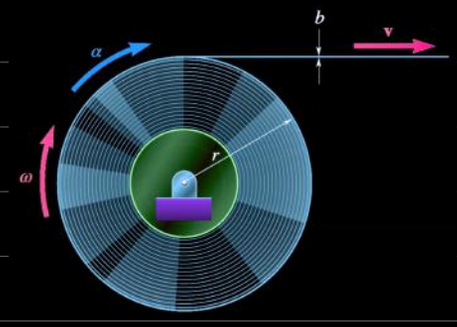
$v = \omega_0 r$, thus $a = \omega_0 \, dr/dt$. Now in each turn the thickness of the spool increased by b thus approximating the spool like a circle, the radius at a general time t is $r = r_0 + \left(\frac{\omega_0 t}{2\pi}\right) \times b$

Thus $\frac{dr}{dt} = \omega_0 b / 2\pi$.

Thus acceleration of the tape, $a = \omega_0 \times \omega_0 b/2\pi = \omega_0^2 b / 2\pi$ Ans.

EXAMPLE:

In a continuous printing process, paper is drawn into the presses at a constant speed v. Denoting by r the radius of the paper roll at any given time and by b the thickness of the paper, derive an expression for the angular acceleration of the paper roll.



Solution: $v = \omega r \Rightarrow \omega = v/r \Rightarrow \frac{d\omega}{dt} = \frac{-1}{r^2}\frac{dr}{dt} \times v$

Also, $r = r_0 - \left(\int \omega \, dt\right) \frac{1}{2\pi} \times b \Rightarrow \frac{dr}{dt} = \frac{-b\omega}{2\pi} = \frac{-b}{2\pi} \cdot \frac{v}{r}$

thus $\alpha = \frac{-1}{r^2} \cdot \frac{-bv}{2\pi r} \times v = \frac{v^2 b}{2\pi r^3}$ Ans.

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Example: A cyclist uses a stationary trainer during the winters to keep in shape. Knowing that she pushes down on her pedal with velocity of 26 in/s that increases at 2 in/sec², determine velocity and acceleration of point D on the bottom of the rear wheel. (Point D is on conveyor)



Solution: Let P: Pedal G: Pedal Gear S: Sprocket then, W: wheel

$$\omega_p = \frac{v_p}{r_p} = \omega_G \text{, also } \omega_G r_G = \omega_S r_S \quad (1)$$

$$\Rightarrow \omega_S = \frac{\omega_G r_G}{r_S} = \frac{v_p}{r_p} \times \frac{r_G}{r_S} \text{ ; } v_D = \omega_S \times r_w \quad (2)$$

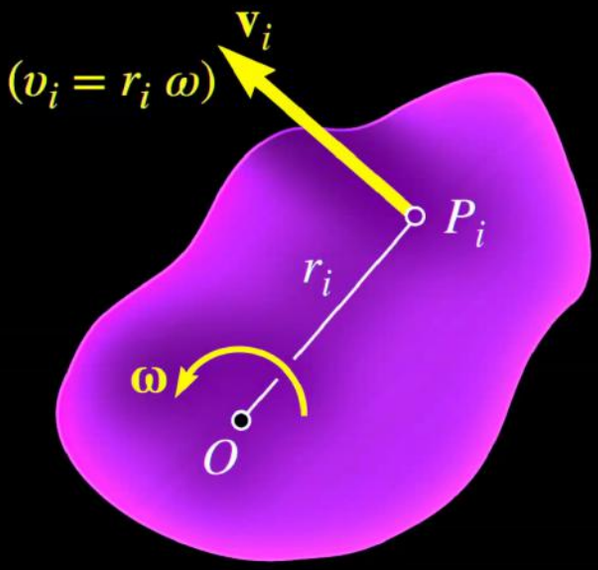
$$\Rightarrow V_D = \frac{V_p}{r_p} \times \frac{r_G}{r_S} \times r_w = \frac{26\ in/sec}{6.5\ in} \times \frac{3.5\ in}{1.75\ in} \times 14\ in = 112\ in/sec$$

$$\text{Now } a_D = \frac{dV_D}{dt} = \frac{a_p}{r_p} \times \frac{r_G}{r_S} \times r_w = \frac{2\ in/sec^2}{6.5\ in} \times \frac{3.5\ in}{1.75\ in} \times 14\ in = 8.61\ in/sec^2$$

→ Since bike is stationary and D is moving, had D been stationary, bike would have appeared to be moving with V_D & a_D forward.

Kinetic Energy In Fixed Axis Rotation:

Consider a rigid body rotating about a fixed axis as shown with an angular velocity ω. If r_i is the ⊥ distance of a point P_i from the axis, its speed is given as: $v_i = \omega r_i$, thus if its mass is m_i, its kinetic energy is given as $KE_i = \frac{1}{2} m_i (\omega r_i)^2$. Thus



the kinetic energy of entire rigid body is given as:

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$KE = \sum KE_i = \frac{1}{2}\omega^2 \sum m_i r_i^2$. The quantity $\sum m_i r_i^2$ is called the moment of inertia of the rigid body about the given fixed axis. Thus:

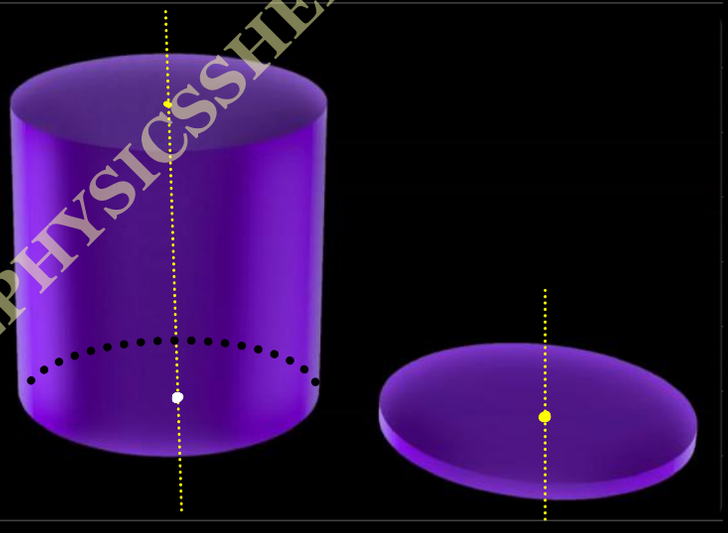
$$\boxed{KE = \frac{1}{2} I_A \omega^2}$$

I_A: Moment of inertia about axis A.

In the upcoming sections, we will be deriving moments of inertia of some standard geometries and also prove and state some important theorems related to moments of inertia of various bodies.

"Pichkoo" Theorem :

If an object is deformed (its particles are moved mentally) in such a way that distance of its particles from the axis is conserved, then the moment of inertia of the object about the axis remains conserved. Thus in the adjacent figure, if we squeeze the cylinder by moving its particles parallel to the axis to form a disc, the moment of inertia of the disc will be same as that of the original cylinder.



→ Axial sweeps are allowed (i.e. moving particles ∥ to axis)

→ Tangential sweeps are allowed (i.e moving particles along circular arcs centred on the axis)

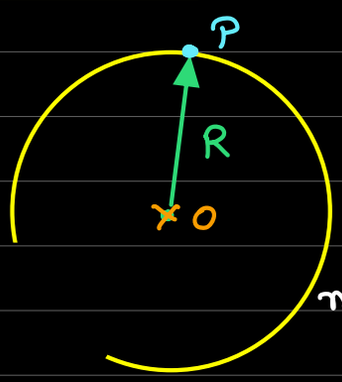
→ Radial sweeps are not allowed (i.e. moving particles means to or away from axis is NOT ALLOWED).

→ Simultaneous axial and tangential sweeps are also allowed.

→ Pichkoo theorem often helps in transforming the geometries with unknown moments of inertia in to one's with known moments of inertia.

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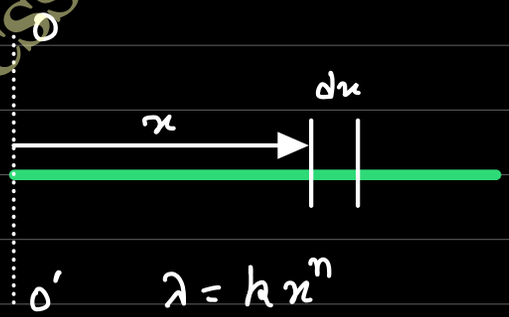
Moment Of Inertia Of A Circular Arc (Axis ⊥ to Plane):



Consider a partial ring of mass m whose moment of inertia is to be found about axis O which is ⊥ to the page and passing through the centre of curvature of the ring. We can now sweep the arc tangentially to collect all the mass at point P. Thus its moment of inertia is as if all mass were located at P (By Pichloo theorem). Thus:

$$\boxed{I = mR^2}$$ ← Also true for complete rings / cylindrical Arc

Moment Of Inertia Of A Rod With Mass Density $\propto x^n$:



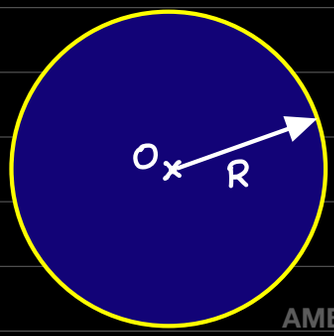
Consider a rod whose linear mass density varies as $\lambda = kx^n$ where x is the distance from the axis OO'. Let its total mass be m and total length be l. We wish to find its moment of inertia about its end point axis OO'. We can do this by integrating elemental moments of inertia. We consider an element of thickness dx at a distance x from the axis. Thus:

$$dI = \lambda dx\, x^2 = k x^{n+2} dx \Rightarrow I = k\int_0^l x^{n+2} dx = \frac{k l^{n+3}}{n+3} \quad (1)$$

Now elemental mass $dm = \lambda dx = kx^n dx$ thus $m = \int_0^l kx^n dx = \frac{k l^{n+1}}{n+1}$ (2)

From (1) & (2) $$\boxed{I = \frac{n+1}{n+3} m l^2}$$

Moment Of Inertia Of A Sector / Disc:



Consider a disc of mass m and radius R. We wish to find its moment of inertia about axis O passing through its centre and ⊥ to its plane. If we tangentially sweep the "dust particles"

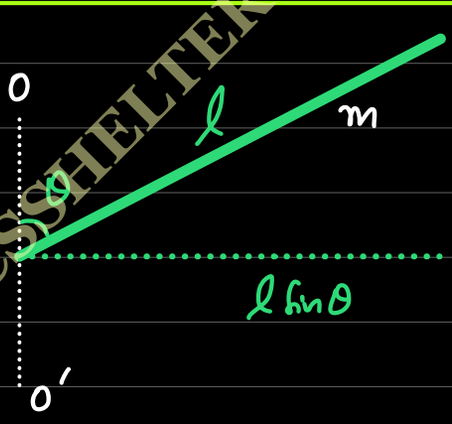
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forming the disc we will have a rod whose mass density will be proportional to r^1 where r is the distance from the axis (This is because particles collected at a distance r will be proportional to $2\pi r$). Thus we will have a rod with $\lambda = kr^1$ accordingly by the result derived in previous section:

$I = mR^2\left(\frac{1+1}{1+3}\right) = \frac{1}{2}mR^2$ thus: $\boxed{I = \frac{1}{2}mR^2}$ ← Also for sectors and cylindrical sectors.

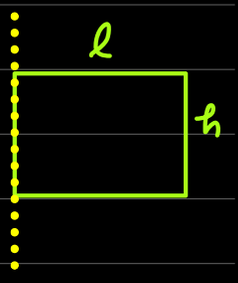
Moment Of Inertia of Uniform Rod About Arbitrary Axis Through End Point:

Consider a uniform rod of mass m and length l whose moment of inertia is to be found about axis OO' which is inclined at an angle θ wrt rod. If we move the particles of rod vertically, we get its projection as $l\sin\theta$, thus transforming it to our standard case of $\lambda = kx^0$, with total length being $l\sin\theta$.



Thus $I = \frac{0+1}{0+3} \times m(l\sin\theta)^2$ i.e $\boxed{I = \frac{ml^2\sin^2\theta}{3}}$

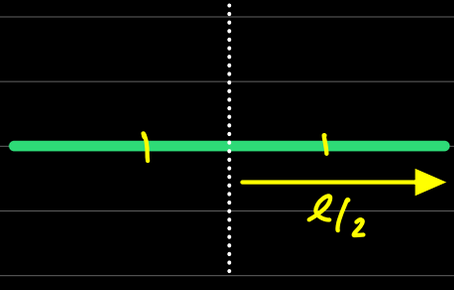
For $\theta = 90°$, $\boxed{I = \frac{ml^2}{3}}$ ← Also true for rectangular lamina



Moment Of Inertia Of A Rod About 1 Bisector:

Here the problem can be divided in to finding moment of inertia of two rods each of length $l/2$. Thus using result of the previous case,

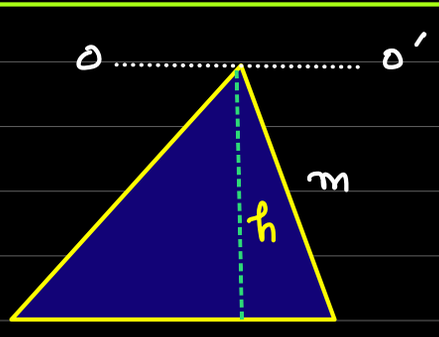
$I = \frac{(m/2)(l/2)^2}{3} \times 2 = \frac{ml^2}{12}$. If instead of $\perp$ the axis were making an angle θ with rod while still passing through mid point then:



$\boxed{I = \frac{ml^2\sin^2\theta}{12}}$

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Moment Of Inertia Of A Triangular Lamina (Axis || to Base Through Apex):

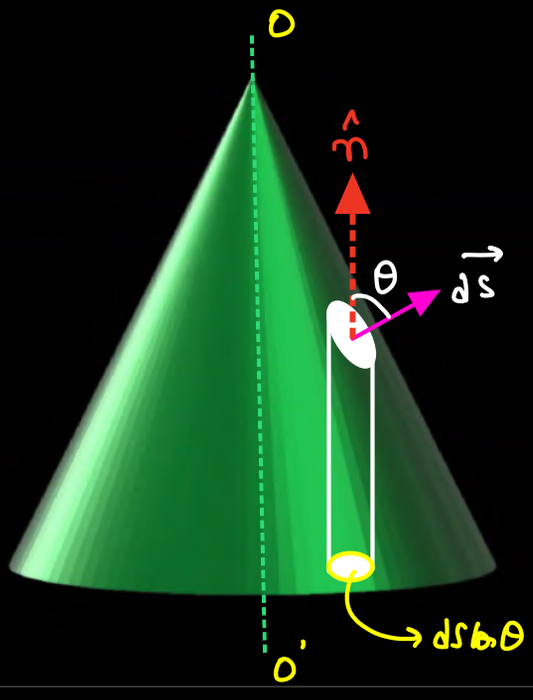


Consider a triangular lamina of mass m and height h whose moment of inertia is to be found about axis OO'. If we squeeze the triangle by sweeping the particles || to OO' we will have a vertical rod of height h with mass density proportional to y' where y is the depth from the axis. Thus by our standard result, $I = mh^2\left(\frac{1+1}{1+3}\right)$

Thus $$\boxed{I = \frac{mh^2}{2}}$$

Moment Of Inertia Of A Conical Shell:

Imagine a conical shell of mass m and radius r whose moment of inertia is to be found about its symmetry axis. Now imagine a patch of area dS whose normal is inclined at angle θ to the vertical. If we sweep these particles vertically, then these particle will occupy an area $dS\cos\theta$ in the base (i.e $\vec{dS}\cdot\hat{n}$). Similarly if we consider such an operation for every patch on the conical shell, its "dust particles" will form a uniform disc on its base. Thus the problem reduces to the previously analyzed case of disc about symmetry axis.

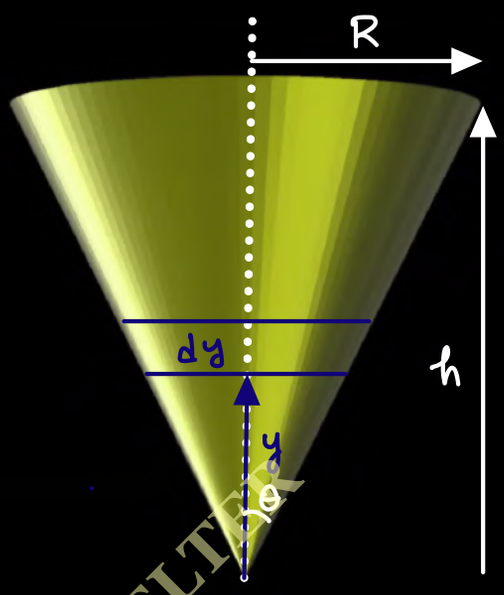


Thus for conical shell, about symmetry axis:

$$\boxed{I = \frac{1}{2}mR^2}$$

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Moment Of Inertia Of A Solid Cone (Symmetry Axis):



Here we divide the cone in to disc elements. Let ρ be the mass density of the cone then for the elemental disc, $dI = \frac{1}{2} dm\, r^2$ where $r = y\tan\theta$

and $dm = \rho \pi y^2 \tan^2\theta \times dy$ thus, $dI = \frac{1}{2}\rho \pi y^4 \tan^4\theta\, dy$, Thus, I:

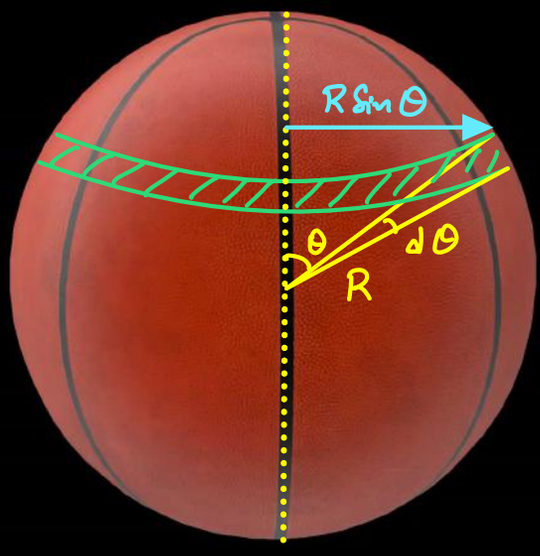
$$I = \frac{1}{2}\rho\pi \tan^4\theta \int_0^h y^4 dy = \frac{1}{2}\rho\pi \tan^4\theta \frac{h^5}{5}$$

Now $\rho = m/\left(\frac{1}{3}\pi R^2 h\right)$ and $\tan\theta = R/h$ substituting these in I:

$$\boxed{I = \frac{3}{10} m R^2}$$

Moment Of Inertia Of A Spherical Shell:

Let the mass of spherical shell be m and its radius be R. Then mass of its ring element between θ & $\theta + d\theta$ is given as $dm = \frac{M}{4\pi R^2} \times 2\pi R\sin\theta \times R\, d\theta$



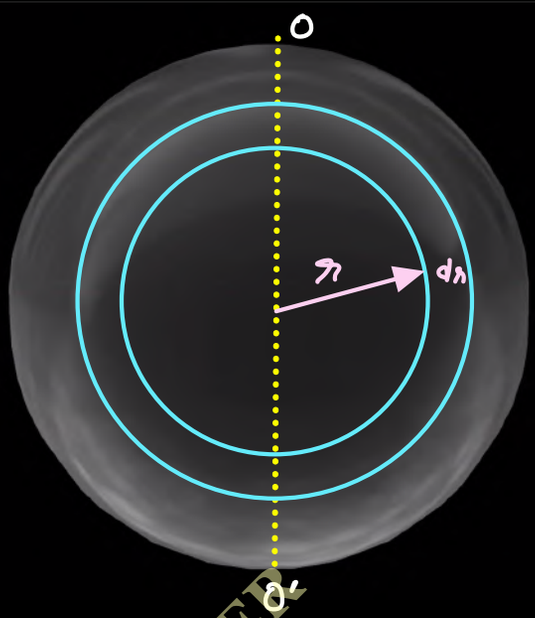
i.e $dm = \frac{M}{2}\sin\theta\, d\theta$. Thus elemental moment of inertia: $dI = dm \times R^2\sin^2\theta$ thus,

$$\int dI = \frac{M}{2}R^2 \int_0^\pi \sin^3\theta\, d\theta = \frac{MR^2}{2}\int_0^\pi (1-\cos^2\theta)\sin\theta\, d\theta$$

Thus $$\boxed{I = \frac{2}{3} M R^2}$$

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Moment Of Inertia Of A Solid Sphere:



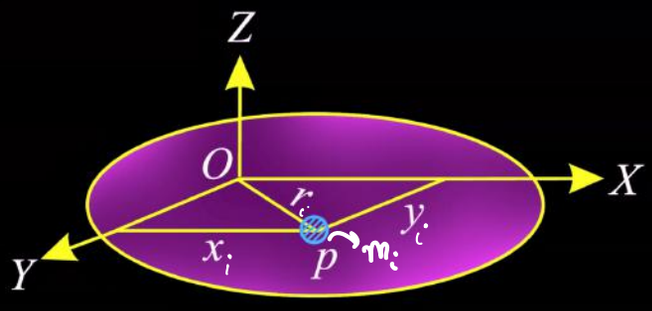
We wish to find moment of inertia of a solid sphere of mass m and radius R about its diameter. We could do it either using disc elements or shell elements. Here we are using shell elements, student is advised to also verify the result using disc elements. Consider a shell element of thickness dr and radius r. Its mass is $dm = \frac{M}{4/3\pi R^3} \times 4\pi r^2 dr = \frac{3M}{R^3} r^2 dr$.

thus $dm = \frac{3M}{R^3} r^2 dr$. Now $dI = \frac{2}{3} dm r^2$, thus:

$$dI = \frac{2}{3} \cdot \frac{3M}{R^3} r^2 dr \times r^2 \quad \text{thus} \quad I = \frac{2M}{R^3} \int_0^R r^4 dr = \frac{2}{5} MR^2$$

thus for solid sphere about diameter: $\boxed{I = \frac{2}{5} mR^2}$

Perpendicular Axis Theorem:



Let I_x and I_y be the moments of inertia about two mutually $\perp$ axes in the plane of a lamina and let I_z be the moment of inertia about an axis $\perp$ to the lamina passing through intersection of X and Y axes then:

$$\boxed{I_z = I_x + I_y}$$

Proof: $I_x = \Sigma m_i y_i^2 \qquad I_y = \Sigma m_i x_i^2$

$$I_z = \Sigma m_i r_i^2 = \Sigma m_i (x_i^2 + y_i^2) = \Sigma m_i x_i^2 + \Sigma m_i y_i^2$$

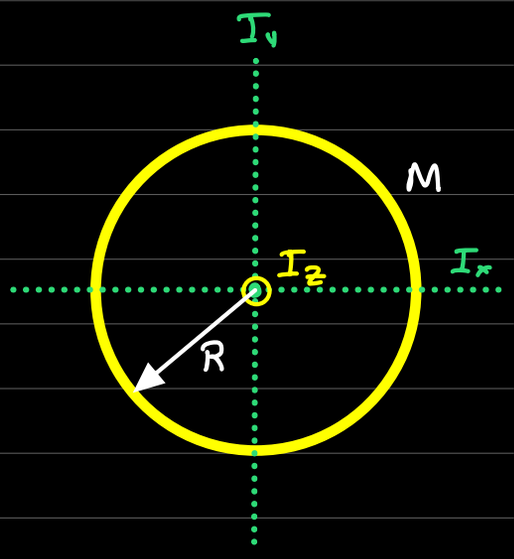
[See figure above].

$\Rightarrow I_z = I_x + I_y$ QED.

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Moment Of Inertia Of Ring About Diameter:

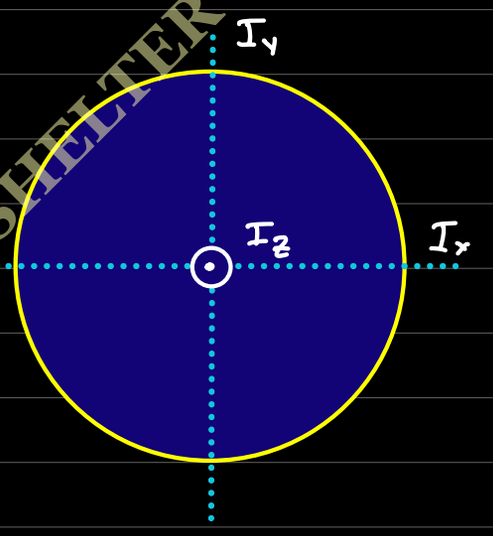
$$I_z = MR^2 = I_x + I_y = 2I_x \Rightarrow \boxed{I_x = \frac{MR^2}{2}}$$



Moment Of Inertia Of Disc About Diameter:

$$I_z = \frac{1}{2}MR^2 = I_x + I_y = 2I_x$$

$$\Rightarrow \boxed{I_x = \frac{1}{4}MR^2}$$



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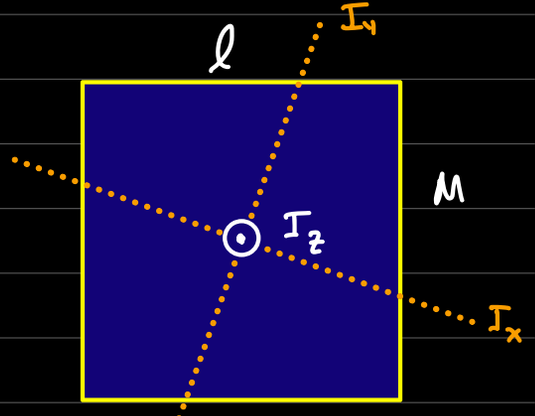
Moment Of Inertia Of A Square Lamina About Arbitrary Centroidal Axis.

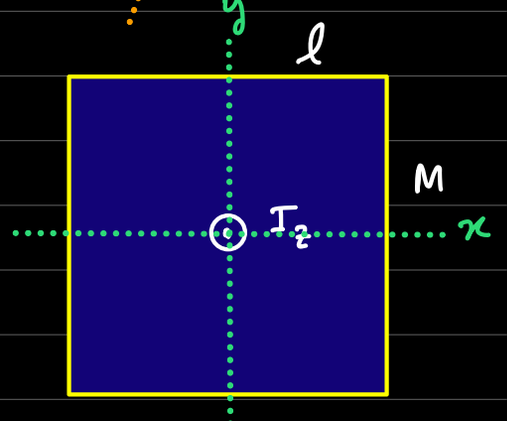
$$I_z = I_x + I_y = 2I_x$$

$$\Rightarrow I_x = I_z/2 \quad (1)$$

Also now refer to the second figure:

$$I_z = I_x + I_y \quad (2)$$





also $I_x = Ml^2/12$ ($\because$ Pichhoo $\longrightarrow$ Rod)

Similarly $I_y = Ml^2/12$ ($\because$ " ")

Thus from (2),

$I_z = Ml^2/6$ (3) using (3) in (1)

$$\boxed{I_x = Ml^2/12}$$ ← Arbitrary centroidal axis (in plane).

AMBARISH SIR

Radius Of Gyration:

The radius of Gyration (k) about any axis for any rigid body is defined by the relation:

$$I = mk^2$$

where m is the mass of the body, I is the moment of inertia and k is the radius of Gyration. Thus

$$k = \sqrt{\frac{I}{m}}$$

Radius of Gyration represents a kind of average distance of particles from the axis for the purpose of finding moment inertia.

By definition of radius of Gyration, we can list out some of the radii of Gyrations as:

Disc	about Symmetry axis	: $R/\sqrt{2}$
"	" Diameter	: $R/2$
Solid Sphere	" "	: $\sqrt{2/5}\,R$
Hollow "	" "	: $\sqrt{2/3}\,R$

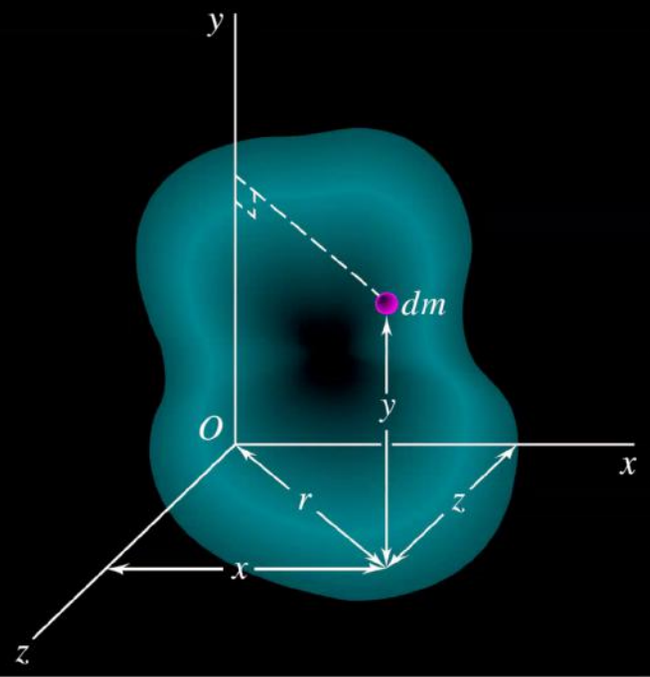
etc.

Moments Of Inertia Wrt Coordinate Axes:

$$I_x = \int (y^2 + z^2)\,dm$$

$$I_y = \int (z^2 + x^2)\,dm$$

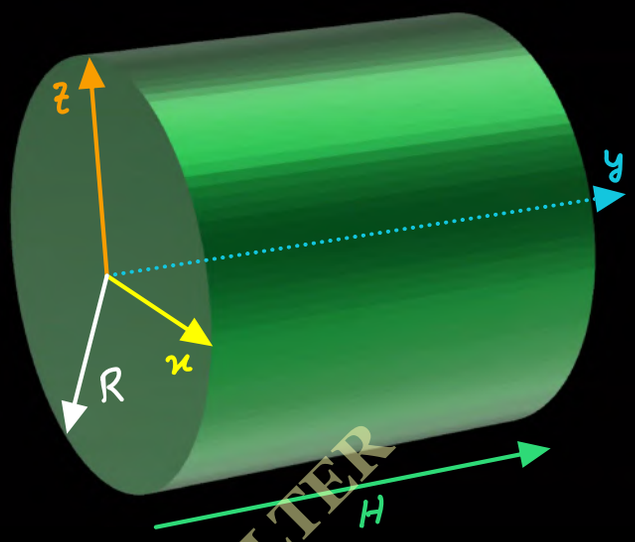
$$I_z = \int (x^2 + y^2)\,dm$$



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Finding Moment Of Inertia By Superposition Of Terms:

Suppose we wish to find moment of inertia of the shown cylinder about z axis. We could of course do it by integrating the moments of inertia of its elemental discs (Along with use of ll axis theorem that we will deal with in an upcoming section). However with a little use of creativity we can convert this problem in to superposition of known problems.



Now $I_z = \sum m_i (x_i^2 + y_i^2) = \sum m_i x_i^2 + \sum m_i y_i^2$ (A)

Now suppose I move the particles in such a way that their x coordinates are maintained but y coordinates are made zero then I will have a disc in x-z plane whose $\sum m_i x_i^2$ is still same as that of my original cylinder. Thus $\sum m_i x_i^2 = mR^2/4$ (1) (∵ It is simply I of a disc about diameter).

Now suppose I move the particles of the body in such a way that I make x and z coordinates of all the particles zero while maintaining their y coordinates then I have a uniform rod along y axis whose $\sum m_i y_i^2$ is same as that of original cylinder. Thus, $\sum m_i y_i^2 = mH^2/3$ (2) (∵ It is simply M.I. of rod about end axis).

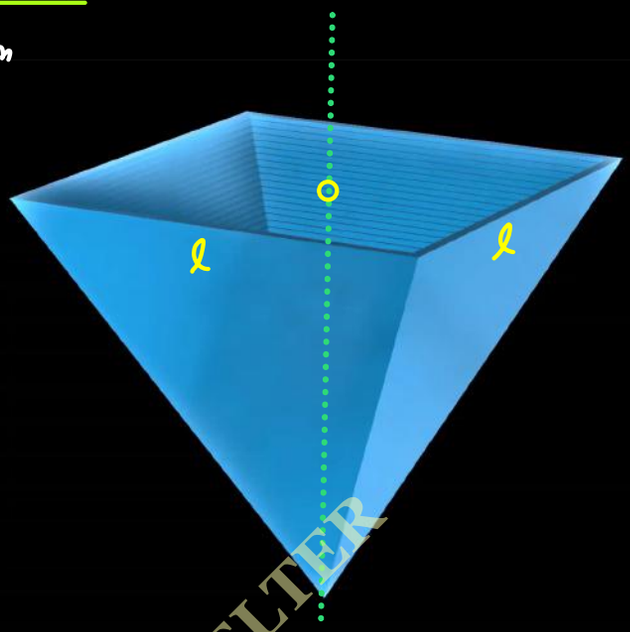
Now from equations (A), (1) and (2),

$$I = \frac{mR^2}{4} + \frac{mH^2}{3} \quad \text{Ans.}$$

→ You can use the same technique to find MI of a conical shell, Pyramid shell or a solid cone / pyramid for M.I. about an axis through apex ll to the base.

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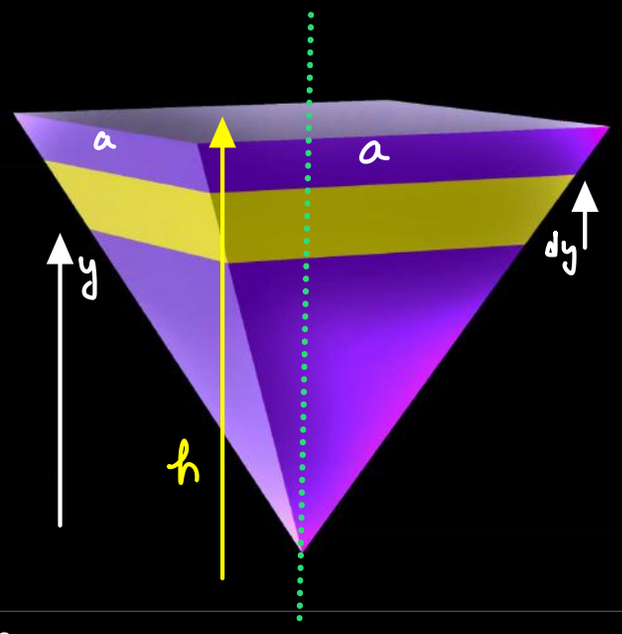
Moment Of Inertia Of A Pyramid Shell (Thin):



Consider a pyramid shell defined by polygon of side l (as shown in the figure). We wish to find its moment of inertia about the dotted axis. Just as with a conical shell, here also we can allow the "dust particles" forming the shell to fall to ground (moving $\parallel$ to axis). Thus here we will have a uniform square lamina whose MI we already found in a previous section. Thus, here $I = ml^2/6$ Ans.

→ The same technique can be applied to any arbitrary pyramid shell to convert the problem in to finding M.I. of a lamina.

M.I. Of A Solid Pyramid (Relation With Corresponding Polygon):



If the moment of inertia of a polygon is mk^2 then the moment of inertia of corresponding solid pyramid is $3/5\ mk^2$ (for example for disc, MI is $1/2\ mR^2$ while for circular pyramid i.e solid cone, M.I. is $\frac{3}{5} \times \frac{1}{2} mR^2$ i.e $\frac{3}{10} mR^2$). This can be shown using some geometry and calculus.

Let formula for polygonal M.I. be kma^2 then for our element, $dI = k\,dm\left(\frac{ay}{h}\right)^2$ also if ρ is mass density, $dm = \rho \times \left(\frac{y}{h} a\right)^2 dy$

(∵ By similar Δs, side of element is $(y/h) \times a$). Thus,

$$dI = \rho \left(\frac{ya}{h}\right)^2 \times \left(\frac{ya}{h}\right)^2 \times k \Rightarrow I = \frac{k\rho}{h^4} a^4 \frac{h^5}{5} = k\rho a^4 h/5 \quad (1)$$

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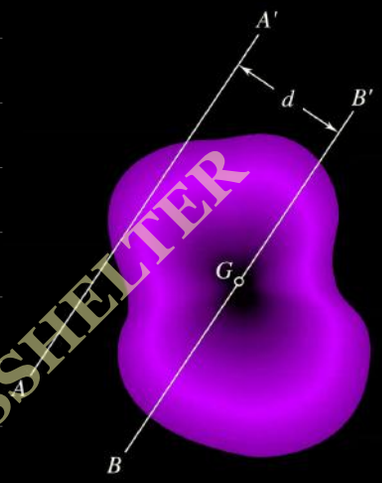
Also mass of the pyramid is $m = \frac{1}{3} \times a^2 h \times \rho$ ②

Using ② in ①, $I = \frac{3}{5} m h a^2 = \frac{3}{5} I_{polygon}$. QED.

Parallel Axis Theorem:

Let $\bar{I}$ be moment of inertia about some centroidal axis (BB' in the figure) and let I be the moment of inertia about an axis parallel to the centroidal axis at a distance d from it then:

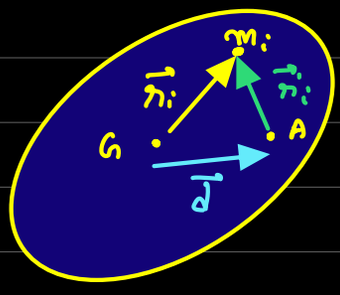
$$\boxed{I = \bar{I} + md^2}$$



We use d to denote the distance between an arbitrary axis AA' and a parallel centroidal axis BB'.

Proof Of ∥ Axis Theorem:

Let $\vec{r}_i$ be perpendicular vector to i^{th} mass (m_i) wrt the centroidal axis and $\vec{r}_i'$ be the ⊥ vector wrt parallel axis (A) then:



$$I = \sum m_i r_i'^2 = \sum m_i (\vec{r}_i' \cdot \vec{r}_i')$$

$$= \sum m_i (-\vec{d} + \vec{r}_i) \cdot (-\vec{d} + \vec{r}_i) = \underbrace{\sum m_i d^2}_{(I)} + \underbrace{\sum m_i r_i^2}_{(II)} - \underbrace{2\vec{d} \cdot \sum m_i \vec{r}_i}_{(III)}$$

But since G is a centroidal axis $\therefore \sum m_i \vec{r}_i / m$ must be zero because it is coordinate of CM in the plane ⊥ to the axis $\therefore$ it must lie on centroidal axis itself. Thus III term in our expression goes to zero and we get:

$$I = \sum m_i r_i^2 + d^2 \sum m_i = \bar{I} + md^2 \quad \text{QED.}$$

→ This shows that out of a set of infinite parallel axes, the least moment of inertia will be about the centroidal axis

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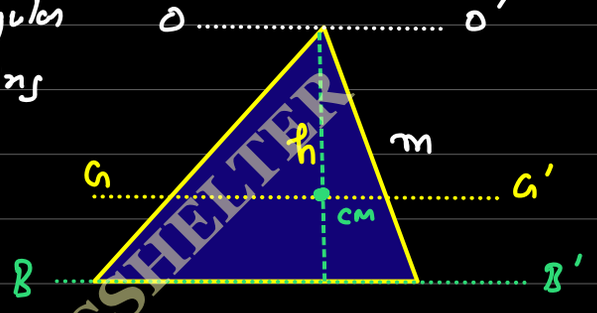
→ (caution!) Parallel axis theorem always involves centroidal axis and can't be used to directly calculate M.I. from one axis to another. For instance if two axes (say A & B) are located at distance r_{AB} from each other and d_A and d_B from centroidal axis then,

$I_A \neq I_B + m r_{AB}^2$ but $I_A = I_B - m d_B^2 + m d_A^2$

→ For a fixed d, I for various axes is same irrespective of location diff.

Moment Of Inertia Of A Triangular Lamina About Base:

Recall that earlier we found I of a triangular lamina about the OO' axis through apex using "Pichhoo" theorem and it came out to be $I_{OO'} = mh^2/2$ now to find



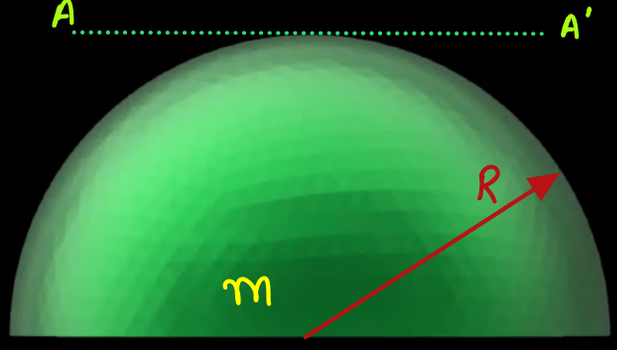
$I_{BB'}$ we can apply // axis theorem.

Accordingly $I_{BB'} = I_{OO'} - m\left(\frac{2h}{3}\right)^2 + m\left(\frac{h}{3}\right)^2$ (∵ CM is at h/3 from base).

Thus $I_{BB'} = \frac{mh^2}{2} - \frac{4}{9}mh^2 + \frac{1}{9}mh^2 = mh^2/6$

$$\boxed{I_B = \frac{mh^2}{6}}$$ ← Moment of inertia of triangular lamina about base.

Example: Find the moment of inertia of solid hemisphere about axis AA'. Mass of hemisphere is m and its radius is R.



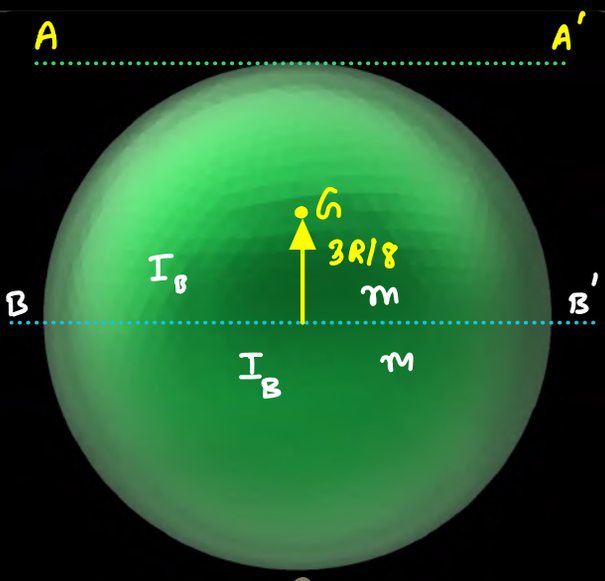
Solution: Here first we will try to find moment of inertia about the diameter of the base using symmetry arguments and then try to use // axis theorem appropriately to find the moment of inertia about the desired axis.

To find MI about diameter of the base, we can complete the solid sphere by adding one more hemisphere at the bottom and then use the standard result for solid sphere as $I = \frac{2}{5}mR^2$.

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Let I_B the moment of inertia of the hemisphere about the base, then:

$$2I_B = \frac{2}{5}(2m)R^2 \Rightarrow I_B = \frac{2}{5}mR^2$$



Now applying ⊥ axis theorem (carefully!)

$$I_A = \frac{2}{5}mR^2 - m\left(\frac{3R}{8}\right)^2 + m\left(\frac{5R}{8}\right)^2$$

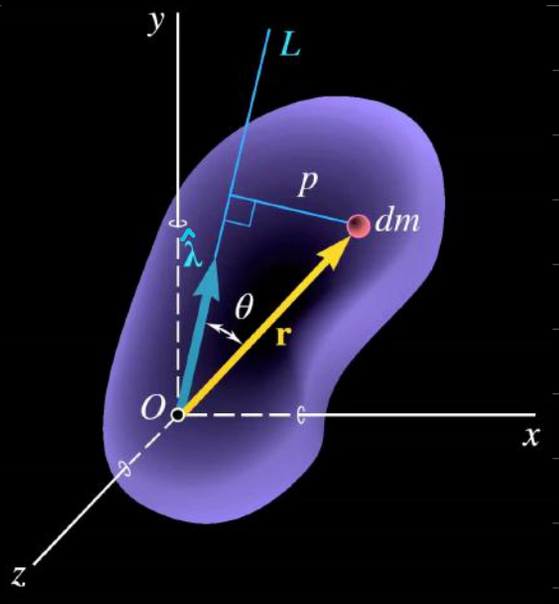
$$= \frac{13}{20}mR^2 \quad \text{thus} \quad I_A = \frac{13}{20}mR^2 \text{ Ans.}$$

→ Pay attention to how not to apply parallel axis theorem. We couldn't simply write $I_A = I_B + mR^2$ (i.e $\frac{7}{5}mR^2$). That would be wrong.

Moment Of Inertia About Axis With Arbitrary Direction Cosines:

Let $\hat{\lambda}$ be a unit vector along an axis passing through origin O such that:

$\hat{\lambda} = \lambda_x \hat{i} + \lambda_y \hat{j} + \lambda_z \hat{k}$ and we wish to find moment of inertia of the rigid body about the axis OL along $\hat{\lambda}$. Let p be the ⊥ distance of an elemental mass dm from this axis, then our required moment of inertia is given as:



$$I_{OL} = \int p^2 dm = \int |\hat{\lambda} \times \vec{r}|^2 dm$$

$$= \int [(\lambda_x y - \lambda_y x)^2 + (\lambda_y z - \lambda_z y)^2 + (\lambda_z x - \lambda_x z)^2]\, dm$$

$$= \lambda_x^2 \int (y^2+z^2)dm + \lambda_y^2 \int (x^2+z^2)dm + \lambda_z^2 \int (x^2+y^2)dm$$

$$- 2\lambda_x\lambda_y \int xy\, dm - 2\lambda_y\lambda_z \int yz\, dm - 2\lambda_z\lambda_x \int xz\, dm \quad (1)$$

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Mass Products Of Inertia

Now we define the products of inertia which involve the last 3 terms in the integrals in equation (1) as:

$$I_{xy} = \int xy\, dm \qquad I_{yz} = \int yz\, dm \quad \text{and} \quad I_{zx} = \int zx\, dm$$

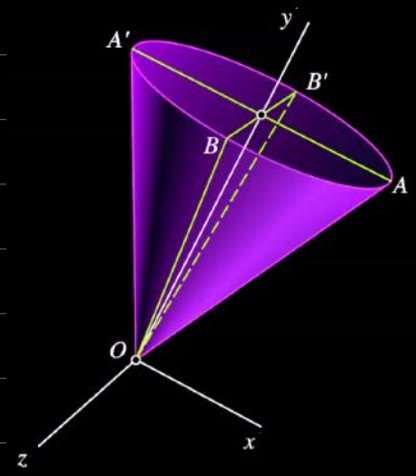
Thus moment of inertia about an arbitrary axis can be written compactly from equation (2) as:

$$I_{OL} = I_x \lambda_x^2 + I_y \lambda_y^2 + I_z \lambda_z^2 - 2\left[I_{xy} \lambda_x \lambda_y + I_{yz} \lambda_y \lambda_z + I_{zx} \lambda_z \lambda_x \right] \quad (3)$$

Principal Axes:

The set of mutually perpendicular x, y and z axes about which all the products of inertia become zero are called principal axes. Principal axes are often very easy to locate. For bodies with (cylindrical) symmetries (like cone or cylinder), One of the axes can be chosen as symmetry axis and other two can be any mutually $\perp$ axes in plane $\perp$ to symmetry axis with origin on symmetry axis. (In fact if any body has two mutually $\perp$ planes of symmetry then the intersection of planes can be chosen as one axis and other two can be mutually $\perp$ axes with origin anywhere on this axis).

For instance in the adjacent figure we have a cone where two mutually $\perp$ planes of symmetry are marked. Accordingly here, x, y and z axes form a set of principal axes. Note that principal axes need not be unique. For example here we could also choose origin at intersection of AA' and BB' or the centroid. The same scheme can be used anywhere on symmetry axis.



A homogeneous cone with elliptical base has two mutually perpendicular planes of symmetry.

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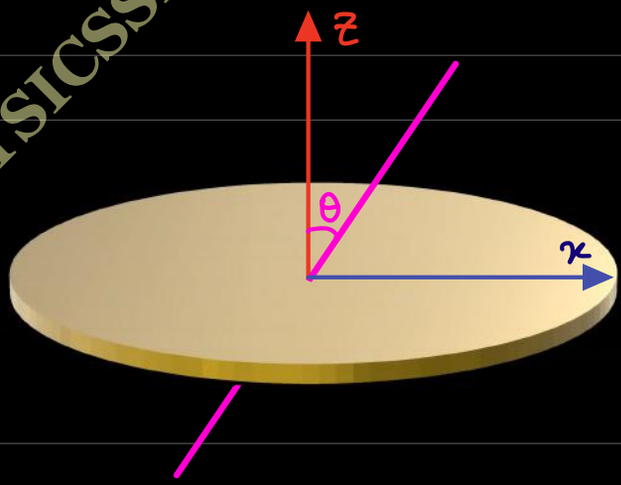
Moment Of Inertia About Arbitrary Axis In Terms Of Principal Moments:

Suppose we are working with principal axes x, y and z (i.e such that I_{xy}, I_{yz} & I_{zx} are each zero), then equation ③ becomes very much simplified and we are left with:

$$I = I_x \lambda_x^2 + I_y \lambda_y^2 + I_z \lambda_z^2 \quad \text{i.e.}$$

$$\boxed{I = I_x \cos^2\alpha + I_y \cos^2\beta + I_z \cos^2\gamma}$$ where $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are direction cosines of the axis along which we wish to find the moment of inertia.

Example: Find the moment of inertia of a thin coin of mass m and radius R about a centroidal axis making an angle θ with its normal (see figure).



Solution: $I = I_x \cos^2\alpha + I_y \cos^2\beta + I_z \cos^2\gamma$

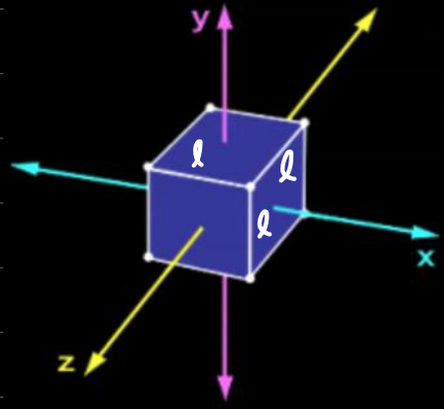
$$= \frac{mR^2}{4}\sin^2\theta + \frac{mR^2}{4}\cos^2 90^\circ + \frac{mR^2}{2}\cos^2\theta$$

$$= \frac{mR^2}{4}\left[\sin^2\theta + 2\cos^2\theta\right] = \frac{mR^2}{4}\left[1 + \cos^2\theta\right] \text{ Ans.}$$

Example: Prove that moment of inertia of a cube about any centroidal axis is $ml^2/6$ irrespective of the direction cosines.

Solution: Consider the cube with centroidal axes shown in the figure. By Pitchboo theorem,

$I_x = I_y = I_z = ml^2/6$ (∵ squeezes to a square).



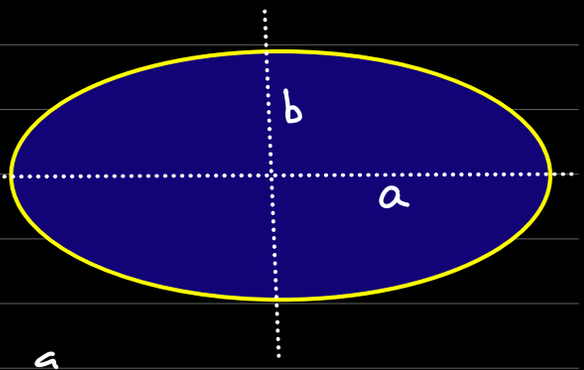
Now:

$$I = I_x \cos^2\alpha + I_y \cos^2\beta + I_z \cos^2\gamma = \frac{ml^2}{6}\left(\cos^2\alpha + \cos^2\beta + \cos^2\gamma\right)$$

But $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ irrespective of direction thus:

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$I = ml^2/6$ irrespective of direction QED.

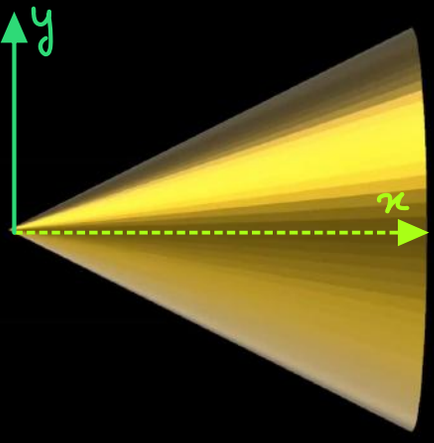


EXAMPLE: Find the moment of inertia of an elliptical lamina about its major axis.

Solution: Recognize that the ellipse is nothing but a circle of radius a, squeezed by a factor of b/a along y axis. Now since moment of inertia is proportional to square of linear dimension, in the squeezing process, it will get scaled by a factor of $(b/a)^2$. Thus required moment of inertia:

$$I = \frac{ma^2}{4} \times \left(\frac{b}{a}\right)^2 = \frac{mb^2}{4} \text{ Ans.}$$

EXAMPLE: Find the moment of inertia of a solid cone of radius R and apex to base distance h about the y axis (see figure).



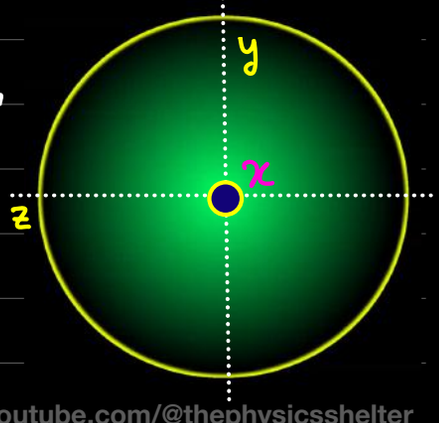
Solution: $I_y = \Sigma m_i(x_i^2 + z_i^2) = \Sigma m_i x_i^2 + \Sigma m_i z_i^2$

Now we will try to evaluate the two terms using superposition. Suppose we squeeze the cone bringing all the particles to x axis maintaining their x coordinates then still $\Sigma m_i x_i^2$ will remain same. This will give us a rod with mass density $\propto x^2$

Thus using the idea of rod simplification, $\Sigma m_i x_i^2 = \left(\frac{2+1}{2+3}\right) m h^2 = \frac{3}{5} m h^2$.

Now suppose we squeeze the cone to make x coordinates of all the particles zero, while maintaining y and z coordinates, $\Sigma m_i z_i^2$ will still remain the same and we will be left with a non uniform disc. Now the challenge is to find the moment of inertia of this non-uniform disc about y axis. Now for this $I_y = I_z = I_x/2$

To find I_x, we stretch the disc back along x axis to form the solid cone again so now I_x for this must be $3/10\, MR^2 \Rightarrow I_y = I_z = \frac{3}{20} mR^2$. Thus we

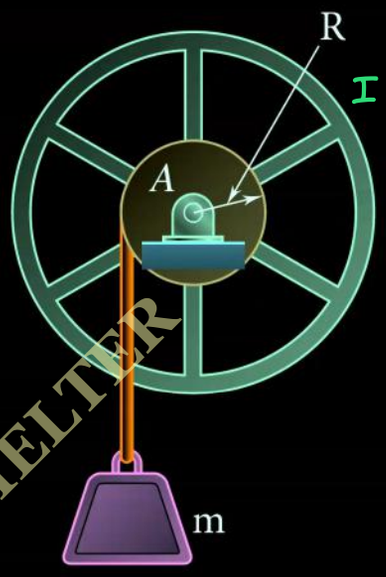


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evaluated our required second term. Accordingly required moment of inertia is $I = \frac{3}{20} mR^2 + \frac{3}{5} mh^2$ Ans.

"Effective Moment Of Inertia" (Single Degree Of Freedom Fixed Axis Systems):

Many times we come across systems in which we have a body rotating about fixed axis connected to other elements through strings / belts and planks etc. The system is said to be of single degree of freedom if the angular position of the rotating member automatically fixes the positions of all the elements in the system. For such systems if ω is the angular velocity of the rotating member then we define effective moment of inertia using kinetic energy as :



$$KE = \frac{1}{2} I_{eff} \omega^2 \quad \text{①}$$

For instance in the figure shown above, the moment of inertia of the wheel is I. Now suppose wheel is rotating with angular velocity ω then velocity of m is ωR and accordingly its KE is $\frac{1}{2} m(\omega R)^2 = \frac{1}{2} mR^2\omega^2$ thus net KE of the system is $KE = \frac{1}{2} I\omega^2 + \frac{1}{2} mR^2\omega^2 = \frac{1}{2} (I + mR^2)\omega^2$

Thus by comparing with equation ①, $I_{eff} = I + mR^2$. Here m is behaving as if it were attached to the rim of the wheel and that gives us a quick way of finding effective moment of inertia.

Finding Angular Acceleration Using Effective Moment Of Inertia:

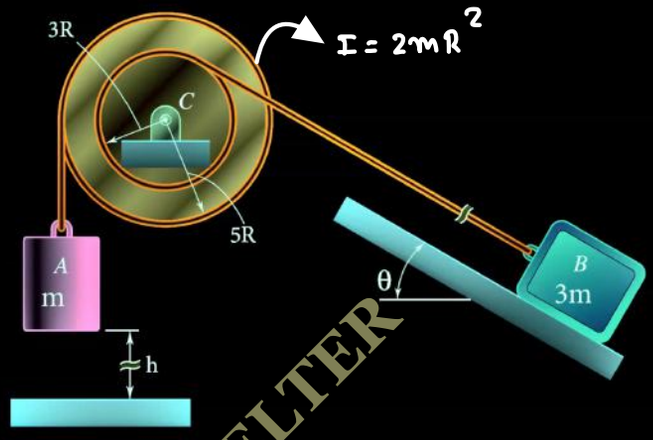
Using work energy theorem, we know that power developed on any system is equal to rate of change of its kinetic energy. Thus:

$$P = \frac{d}{dt}\left[\frac{1}{2} I_{eff} \omega^2\right] = I_{eff}\, \omega \frac{d\omega}{dt} = I_{eff}\, \omega \alpha$$

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Thus $\boxed{\alpha = \frac{P}{I_{eff}\,\omega}}$ In most problems, the power developed will be proportional to first power of ω and thus ω will cancel out in the quotient on the RHS.

EXAMPLE: In the shown system θ is small enough so that block of mass m moves downwards. Find (a) Angular acceleration of the wheel (b) Accelenation of m (c) Tension in the string connected to m (d) Time taken by m to hit the ground.



Solution: $I_{eff} = 2mR^2 + 3m(3R)^2 + m(5R)^2 = 54mR^2$

Now if wheel rotates anticlockwise with angular velocity ω, Power developed by gravitational forces $P = mg(5\omega R) - 3mg\sin\theta(3\omega R)$

$\Rightarrow P = mg\omega R[5 - 9\sin\theta)$, thus, $\alpha = P/I_{eff}\omega = mg\omega R(5 - 9\sin\theta)/54mR^2$

$\Rightarrow \alpha = \frac{g}{54R}[5 - 9\sin\theta)$ Ans. (Thus $\sin\theta$ should be less than 5/9).

Now accelenation of m $a_m = \alpha \times 5R = \frac{5g}{54}[5 - 9\sin\theta]$ Ans.

For m, $mg - T = ma_m \Rightarrow T_m = mg - \frac{5mg}{54}[5 - 9\sin\theta]$

$\Rightarrow T_m = \frac{mg}{54}[29 + 45\sin\theta]$ Ans

Now if t_0 is the time to hit the ground, $\frac{1}{2}a_m t_0^2 = h$

$\Rightarrow t_0 = \sqrt{\frac{2h}{a_m}}$ Ans.

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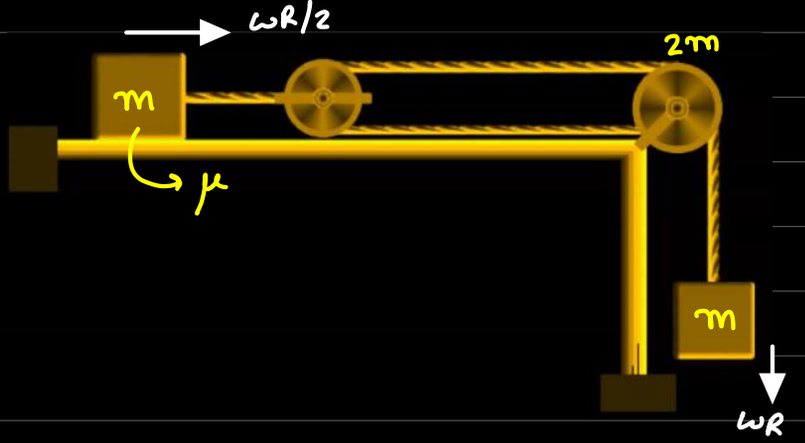
Power Developed By A Torque In Fixed Axis Rotation:

Recall that power developed by a force is $\vec{F}\cdot\vec{v}$, now in case of fixed axis rotation, $\vec{v} = \vec{\omega}\times\vec{r}$ as proved in a previous section. Thus power developed by a force can be written as $P = \vec{F}\cdot(\vec{\omega}\times\vec{r})$. This is scalar triple product of $\vec{F}$, $\vec{\omega}$ and $\vec{r}$ which represents volume of a parallelopipe defined by these three vectors. Thus we can also write this power as

$P = (\vec{r}\times\vec{F})\cdot\vec{\omega}$, thus Power : $\boxed{P = \vec{\tau}\cdot\vec{\omega}}$

Note that $\vec{r}\times\vec{F}$ is torque about a point on the axis, however $\vec{\omega}$ is along the axis so components of torque $\perp$ to the axis won't matter in the dot product. All that matters is the component of torque along the axis. Thus we can also write power in terms of torque about the axis as $\boxed{P = \pm\tau\omega}$
The sign +/- depends on whether $\vec{\tau}$ and $\vec{\omega}$ are parallel or anti-parallel.

Example: Consider the system in adjacent figure. Here the left pulley is massless. Find the angular acceleration of the fixed pulley (It's mass is $2m$ and radius is R). (Pulleys are like solid discs).



Solution: If the fixed pulley rotates by ω, the lower block goes by a velocity ωR while the upper block goes by a velocity $\omega R/2$. Thus their effective moments of inertia become mR^2 and $mR^2/4$. Total effective inertia : $I_{eff} = \frac{1}{2}(2m)R^2 + \frac{mR^2}{4} + mR^2 = \frac{9}{4}mR^2$.

Now Power $P = mg(\omega R) - \mu mg(\omega R/2) = mg\omega R[1 - \mu/2]$

Thus $\alpha = P/I_{eff}\omega = \frac{4g}{9R}[1-\mu/2]$ Ans.

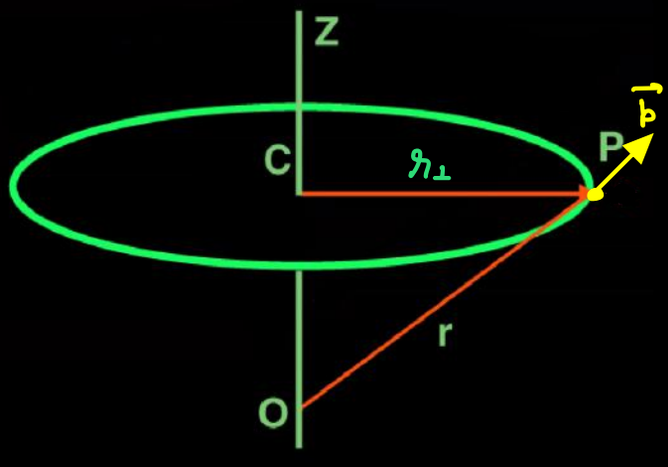
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→ Think: What if left pulley were also massive?

Solution: We know that KE of any system is $\frac{1}{2}mv_{cm}^2 + K_0$ where K_0 is energy as seen from CM. From CM pulley would appear to be in fixed axis rotation. If its radius were r, its angular velocity would be $(v_{top} - v_{bottom})/2r$, since topmost point will appear to be rotating about bottommost point with an angular velocity equal to that of the pulley, thus if ω' is its angular velocity, $\omega' = (\omega R - 0)/2r$.

Angular Momentum About An Axis:

Just as we distinguished between torque about a point and torque about an axis, in the same manner we can also talk about angular momentum about an axis. The angular momentum about an axis is simply component of angular momentum about a point on the axis along the axis (vector component). Thus if $\hat{n}$ is the unit vector along an axis and $\vec{r}$ is position vector of a particle relative to a point on the axis then its angular momentum about the axis is simply $\vec{L} = [(\vec{r} \times \vec{p}) \cdot \hat{n}]\hat{n}$. Magnitude wise we can write angular momentum about an axis as $L = r_\perp p = r p_\perp$ just as we did for torque. Here $r_\perp$ is the perpendicular distance of the line of motion from the axis.



(One can readily infer why $(\vec{r} \times \vec{p}) \cdot \hat{n}$ will be $r_\perp \times p$ from adjacent figure.

$$(\vec{OP} \times \vec{p}) \cdot \hat{k} = [(\vec{OC} + \vec{CP}) \times \vec{p}] \cdot \hat{k}$$

$= (\vec{OC} \times \vec{p}) \cdot \hat{k} + (\vec{CP} \times \vec{p}) \cdot \hat{k}$, Now $\vec{OC} \times \vec{p}$ is $\perp$ to $\hat{k}$ thus first term here goes to zero and thus torque about axis is simply $\vec{r}_\perp \times \vec{p}$.

→ Even though angular momentum about an axis is a vector quantity, for 2D rotation we will be able to treat it like a signed scalar depending on whether its sense is clockwise or

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anti-clockwise.

Relation Between Torque & Angular Momentum About An Axis:

Recall that we proved for any system of particles that $\vec{\tau}_{ext_O} = \frac{d\vec{L}_O}{dt}$

i.e net torque of external forces about any point O is equal to rate of change of angular momentum about same point O. Now let $\hat{n}$ be the unit vector along some axis A passing through point O then taking dot product on both sides:

$$\vec{\tau}_{ext_O} \cdot \hat{n} = \frac{d}{dt}\left[\vec{L}_O \cdot \hat{n}\right]$$ (Note that $\hat{n}$ is a constant vector so we could put it inside the bracket)

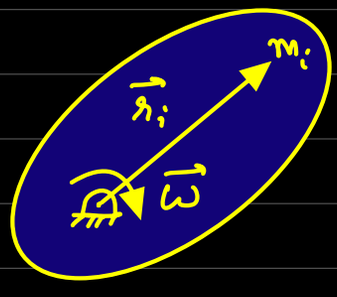
Thus:

$$\left[\vec{\tau}_{ext_O} \cdot \hat{n}\right]\hat{n} = \left(\frac{d}{dt}\left[\vec{L}_O \cdot \hat{n}\right]\right)\hat{n}$$ or simply.

$$\vec{\tau}_{ext_A} = \frac{d\vec{L}_A}{dt}$$ where $\vec{\tau}_{ext_A}$ represents torque of external forces about axis A and L_A represents angular momentum about the same axis.

Angular Momentum Of A Rigid Body About A Fixed Axis:

Let a rigid body be rotating with an angular velocity $\vec{\omega}$ about some fixed axis. Then angular momentum of the ith particle is given as:



$\vec{L}_i = \vec{r}_i \times m_i(\vec{\omega} \times \vec{r}_i)$ where m_i is the mass of ith particle and $\vec{r}_i$ is the $\perp$ vector from axis to the particle. Now $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a}\cdot\vec{c}) - \vec{c}(\vec{a}\cdot\vec{b})$, thus:

$\vec{L}_i = m_i\left[\vec{\omega}(\vec{r}_i \cdot \vec{r}_i) - \vec{r}_i(\vec{\omega}\cdot\vec{r}_i)\right]$, but $\vec{\omega}\cdot\vec{r}_i = 0$ ($\because \perp$).

thus:

$\vec{L}_i = \vec{\omega}\, m_i r_i^2$, now for complete rigid body:

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$\vec{L} = \sum \vec{L}_i = \vec{\omega} \sum m_i r_i^2 = I\vec{\omega}$, thus angular momentum of a rigid body about fixed axis is simply $\vec{L} = I\vec{\omega}$.

Relation Between Torque And Angular Momentum About Fixed Axis:

$\vec{\tau}_A = \frac{d\vec{L}_A}{dt} = \frac{d}{dt}[I_A\vec{\omega}] = I_A\frac{d\vec{\omega}}{dt} = I_A\vec{\alpha}$ thus we have the relation: $\boxed{\vec{\tau}_A = I_A\vec{\alpha}}$

→ Even though this is a vector equation, for 2D rotation, we will treat both torque and angular acceleration as signed scalars depending on whether they are clockwise or anti-clockwise.

Example:

A uniform solid cylinder of mass m can rotate freely about its fixed axis, which is kept horizontal. A particle of mass m_o hangs from the end of a light string which is tightly wound round the cylinder. When system is allowed to move, acceleration with which the particle descends



(A) $\frac{mg}{m+2m_o}$ (B) $\frac{2m_og}{m+2m_o}$ (C) $\frac{m_og}{m+m_o}$ (D) $\frac{2mg}{m+2m_o}$

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Solution: Let T be the tension in the string then by torque equation $TR = \frac{1}{2}mR^2\alpha$ ① ($\because \tau = I\alpha$ for pulley)

Now for point mass, $m_og - T = m_oR\alpha$ ($\because a = R\alpha$ due to no slipping).

Now eliminating T,

$$m_ogR = \left(m_oR^2 + \frac{1}{2}mR^2\right)\alpha \Rightarrow \alpha = \frac{g}{R}\left[\frac{m_o}{m_o + \frac{1}{2}m}\right]$$ Ans.

(B)

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"Effective Torque" Single DOF Fixed Axis Rotation:

When objects are connected to a wheel rotating about fixed axis through strings / planks etc, if external forces are developing powers on them, then torque of external forces may be taken as $\tau_{ext} = P/\omega$, where P is the power and ω is the angular speed of the wheel.

Now angular acceleration may be directly written as:

$$\alpha = \tau_{ext} / I_{eff}$$ where I_{eff} is effective moment of inertia.

Example: Solve the previous problem using effective torque and effective moment of inertia

A uniform solid cylinder of mass m can rotate freely about its fixed axis, which is kept horizontal. A particle of mass m_o hangs from the end of a light string which is tightly wound round the cylinder. When system is allowed to move, acceleration with which the particle descends



(A) $\frac{mg}{m+2m_o}$ (B) $\frac{2m_og}{m+2m_o}$ (C) $\frac{m_og}{m+m_o}$ (D) $\frac{2mg}{m+2m_o}$

Solution: $I_{eff} = \frac{1}{2}mR^2 + m_oR^2$, $\tau_{ext} = m_ogR$, thus,

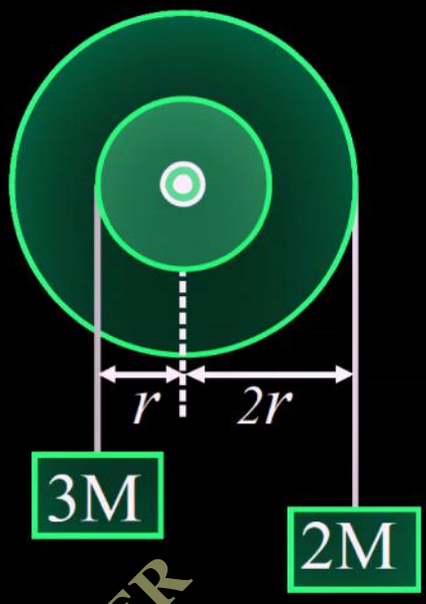
$$\alpha = \tau_{ext}/I_{eff} = m_ogR/\left(\frac{1}{2}mR^2 + m_oR^2\right) = 2m_og/R(2m_o+m)$$

→ Alternatively we could also do it using Power method:

$$\alpha = \frac{P}{I_{eff}\omega} = \frac{m_ogR}{\left(\frac{1}{2}mR^2 + m_oR^2\right)}.$$

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EXAMPLE: For the shown system find the angular acceleration of the pulley assuming it to be massless and perfectly rough. Also find acceleration of 3M.



Solution: $I_{eff} = 3M(r)^2 + 2M(2r)^2 = 11Mr^2$

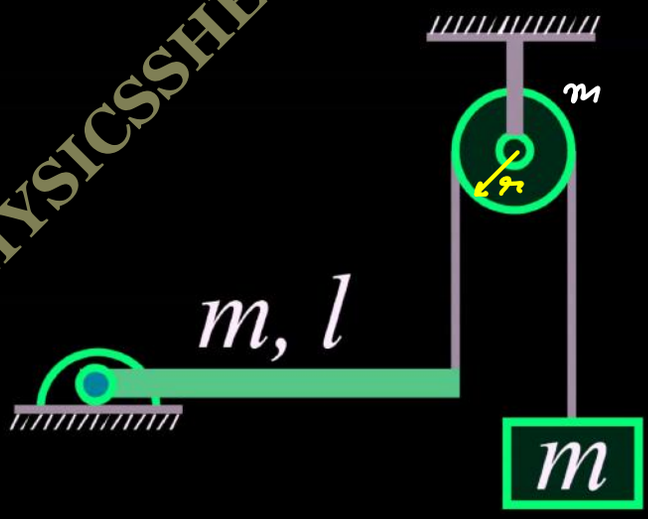
$\tau_{eff} = 2Mg \times 2r - 3Mg \times r = Mgr$

Thus,

$\alpha = \tau_{ext}/I_{eff} = g/11r$ Ans.

Acceleration of 3m: $\alpha \times r = g/11$ Ans.

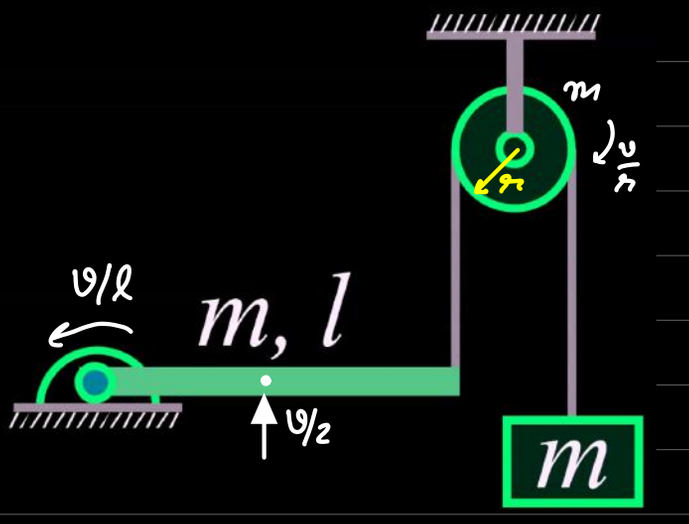
EXAMPLE: In the shown system if $l = 4r$ and mass of pulley is also m and the thread does not slip. Find the acceleration of m.



Solution: If block moves down at velocity v, then angular velocity of pulley is v/r and that of the rod is v/l and the speed of it's mid point is $v/2$ by constraint.

Now power developed by gravity must be equal to rate of change of KE of the system. Thus:

$$(mgv - mg\,v/2) = \frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}\cdot\frac{1}{2}mr^2\left(\frac{v}{r}\right)^2 + \frac{1}{2}\left(\frac{ml^2}{3}\right)\left(\frac{v}{r}\right)^2\right]$$

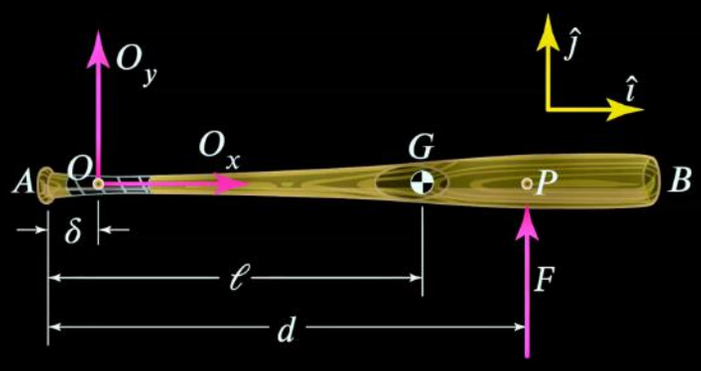


$$\Rightarrow \frac{1}{2}mgv = \frac{d}{dt}\left[v^2\left[\frac{1}{2}m + \frac{1}{4}m + \frac{8}{3}m\right]\right]$$

$$\Rightarrow \frac{1}{2}mgv = 2v\frac{dv}{dt}\left[\frac{41}{12}m\right] \Rightarrow a = \frac{3g}{41}$$ Ans.

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Example: A baseball bat is pivoted on a frictionless horizontal table at point O. Its centre of mass is at G and its moment of inertia about O is I. Find the location of a point P on the bat such that if a horizontal force is applied there, the hinge reaction at O is zero. (Mass is m).



Solution: Using $\tau = I\alpha$, $F(d-\delta) = I\alpha$ ①

Also since hinge reaction is zero, applying $F_{net} = ma_{cm}$:

$F = m(\ell-\delta)\alpha$ ② Now ①/② gives

$$d-\delta = \frac{I}{m(\ell-\delta)} \Rightarrow d = \frac{I}{m(\ell-\delta)} + \delta \text{ Ans.}$$

This point is also called Centre of Percussion or Sweet Point.

→ If the player is holding the bat at O and hits the ball at point P, his hand experiences negligible reaction and a good shot is achieved without an impulse on the hand.

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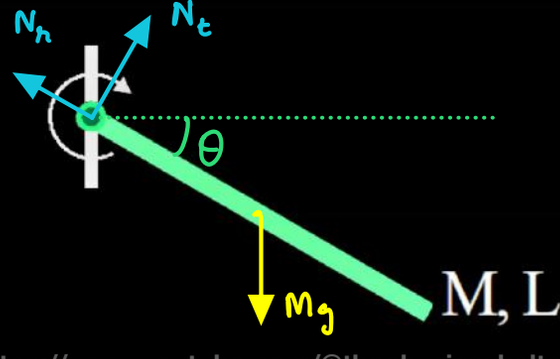
Example: A rod of mass M and length L is released on a frictionless pivot from a horizontal position. After it has turned through an angle θ, find (a) Its angular velocity (b) Components of hinge reaction along the rod and perpendicular to it.



Solution: By mechanical energy conservation:

$$Mg\frac{L}{2}\sin\theta = \frac{1}{2}\frac{ML^2}{3}\omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{3g\sin\theta}{L}} \text{ Ans.}$$



Now Angular acceleration: $Mg\frac{L}{2}\cos\theta = \frac{ML^2}{3}\alpha \Rightarrow \alpha = \frac{3g\cos\theta}{2L}$

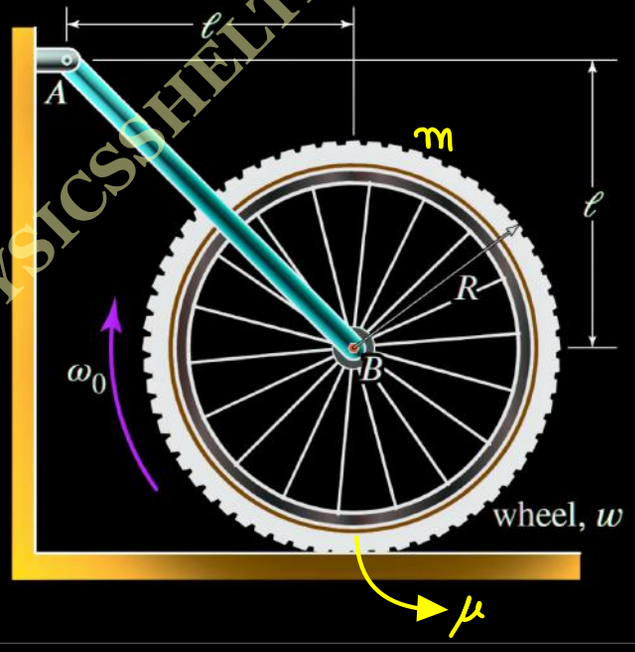
Now apply $\Sigma \vec{F}_{ext} = M\vec{a}_{cm}$ along and $\perp$ to the rod:

$Mg\cos\theta - N_t = M\frac{L}{2}\alpha \Rightarrow N_t = Mg\cos\theta - M\frac{L}{2}\times\frac{3g}{2L}\cos\theta = Mg\cos\theta/4$ Ans.

Similarly along the rod: $N_n - Mg\sin\theta = M\omega^2\frac{L}{2}$

$\Rightarrow N_n = Mg\sin\theta + M\times\frac{3g\sin\theta}{2} = \frac{5}{2}Mg\sin\theta$ Ans.

EXAMPLE: A bicycle wheel is pinned to a wall using a massless rod as shown. If the moment of inertia of the wheel is I, (a) how much time will it take for it to come to rest (b) How many revolutions has it made by then



Solution: Let T be the tension in the rod and N be the normal reaction of the ground, then by horizontal force balance,

$T\cos 45^\circ = \mu N$ ① and

$T\sin 45^\circ + N = mg$ ② (∵ Vertical balance)

From ① & ②, $T = \frac{\sqrt{2}\mu}{1+\mu}mg$ and $N = \frac{mg}{1+\mu}$

Now let the wheel come to rest in n revolutions then by work energy theorem:

$$\mu N \times (2\pi R)n = \frac{1}{2}I\omega_0^2 \Rightarrow n = \frac{I\omega_0^2}{4\pi R\mu N} = \frac{I\omega_0^2(1+\mu)}{4\pi R\mu mg}$$ Ans.

Now Angle turned: $\theta = n\times 2\pi = \left(\frac{\omega_0+0}{2}\right)\times t_0$ (where $t_0 \rightarrow$ stopping time)

$$\Rightarrow t_0 = \frac{2n\pi}{\omega_0} = \frac{I\omega_0(1+\mu)}{\mu mgR}$$ Ans.

(Alternatively we could use torque approach to find α and proceed from there).

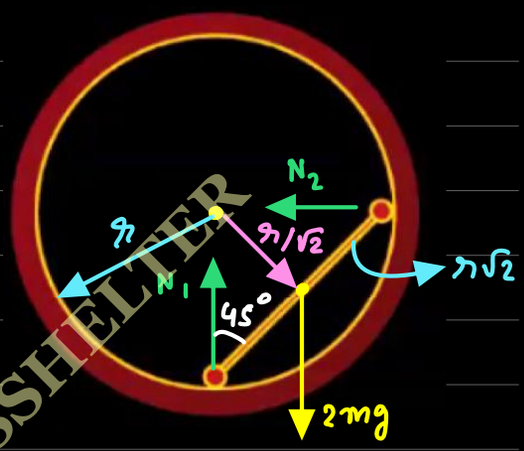
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Example: Pathfinder: Build 27:



Inside a fixed frictionless spherical cavity of radius r, a dumbbell is held touching the bottom of the cavity at its lower end. The dumbbell consists of two identical small balls each of mass m affixed at the ends of a light rod of length $r\sqrt{2}$. What will the force of interaction between the lower ball and the cavity be immediately after the dumbbell is released?

Solution: Notice that even though here fixed axis is not visible, here the entire rod rotates about centre of the cavity as if attached by massless sticks. Also notice that torque of normal reactions about centre of rotation is zero. Now using $\tau = I\alpha$ about axis of rotation:



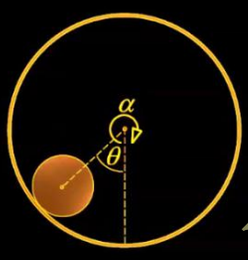
$$2mg\frac{r}{2} = 2mr^2\alpha \Rightarrow \alpha = \frac{g}{2r}$$

Now Using $F_{ext} = ma_{cm}$ in vertical direction:

$$2mg - N_1 = 2m\left(\frac{r}{\sqrt{2}}\alpha \sin 45^\circ\right) = mr\alpha = mg/2$$

$$\Rightarrow \quad N_1 = \frac{3}{2}mg \quad \text{Ans.}$$

Example: Pathfinder: Build 29:



A uniform cylinder is held motionless in a horizontal stationary pipe at an angular position $\theta = 30^\circ$ touching inner surface of the pipe. Radii of the cylinder and the pipe are $r = 0.2$ m and $R = 2.0$ m respectively. The pipe is now made to rotate about its fixed horizontal axis with a constant angular acceleration and the cylinder is released. If the axis of the cylinder remains motionless, find the angular acceleration of the pipe.

Solution: Let angular acceleration of the cylinder be α'

At contact point $\because$ there is no slipping $\therefore$ $r\alpha' = R\alpha$ ①

Now by force balance along friction:

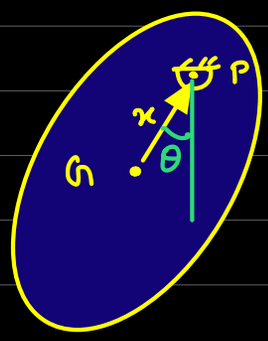
$$f = mg\sin\theta \text{ ②} \qquad (f: \text{friction}).$$

Also by $\tau = I\alpha$, $fr = \frac{1}{2}mr^2 \times \alpha'$ ③ Solving ①, ② & ③:

$$\alpha' = \frac{2g\sin\theta}{r}, \quad \alpha = 2g\sin\theta/R \quad \text{Ans.}$$

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EXAMPLE: A rigid body with radius of gyration k about centroidal axis is to be pivoted at some distance from G so as to have maximum angular acceleration under its own weight. Find the pivot distance x required to achieve the objective.



Solution: $mgx \sin\theta = m(k^2 + x^2)\alpha \quad (\because I = mk^2 + mx^2)$.

$\Rightarrow \alpha = g \sin\theta \left(\frac{x}{k^2+x^2}\right)$, for max α, we must minimize $\left(\frac{k^2+x^2}{x}\right)$

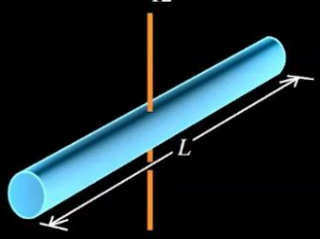
i.e minimize $\frac{k^2}{x} + x$, using $AM \geq GM$, min value is k for $x = k$.

Thus maximum angular acceleration occurs for $x = k$ i.e pivoting at a distance equal to radius of gyration about centroidal axis.

TABLE OF MOMENTS OF INERTIA:

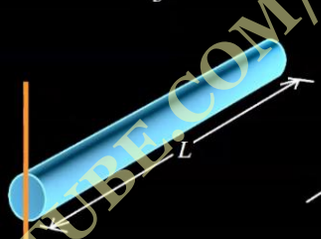
(a) Slender rod, axis through center

$I = \frac{1}{12}ML^2$



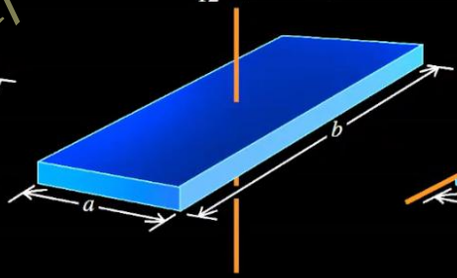
(b) Slender rod, axis through one end

$I = \frac{1}{3}ML^2$



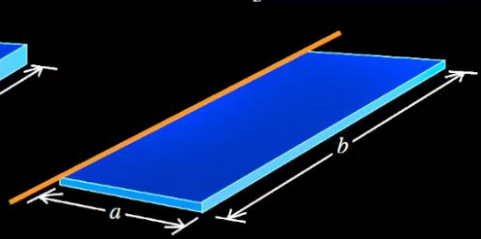
(c) Rectangular plate, axis through center

$I = \frac{1}{12}M(a^2 + b^2)$



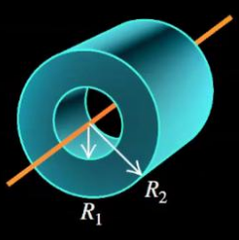
(d) Thin rectangular plate, axis along edge

$I = \frac{1}{3}Ma^2$



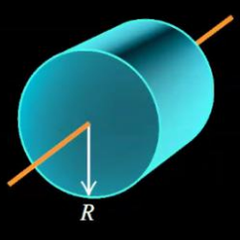
(e) Hollow cylinder

$I = \frac{1}{2}M(R_1^2 + R_2^2)$



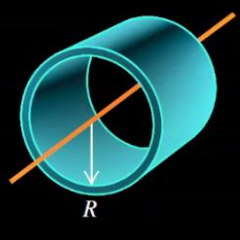
(f) Solid cylinder

$I = \frac{1}{2}MR^2$



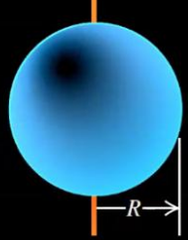
(g) Thin-walled hollow cylinder

$I = MR^2$



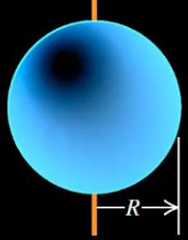
(h) Solid sphere

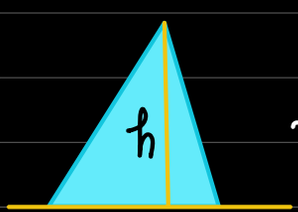
$I = \frac{2}{5}MR^2$



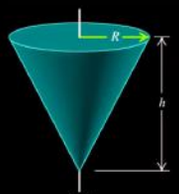
(i) Thin-walled hollow sphere

$I = \frac{2}{3}MR^2$





$mh^2/6$



$3/10\ mR^2$

Hollow Cone: $1/2\ mR^2$

Pyramid: $\frac{3}{5}mh^2$

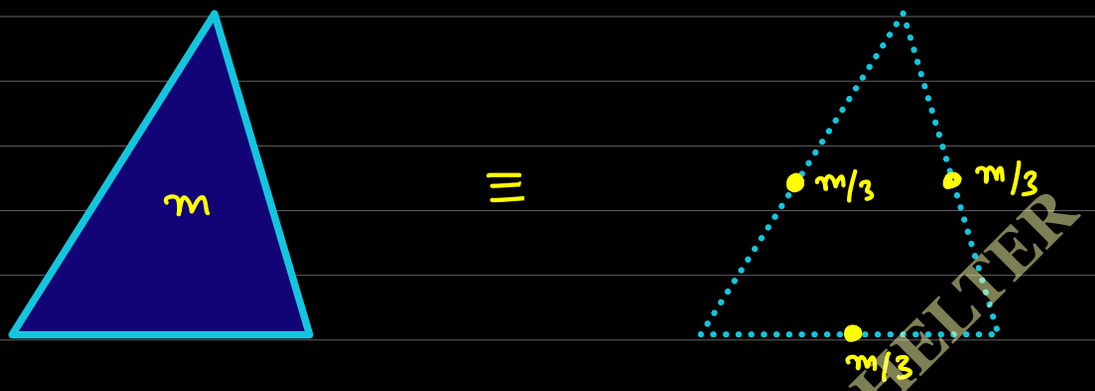
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EQUIMOMENTAL SYSTEMS: The systems which have same net mass and same moment of inertia about any arbitrary axis are called equimomental systems. One of the equimomental systems especially useful is equivalence of arbitrary triangular lamina to three point masses at mid points of its sides as shown in the following figure.



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